Seismic Noise

Han Xiao

GFZ Helmholtz Centre for Geosciences

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Outline

1. Introduction of seismic noise

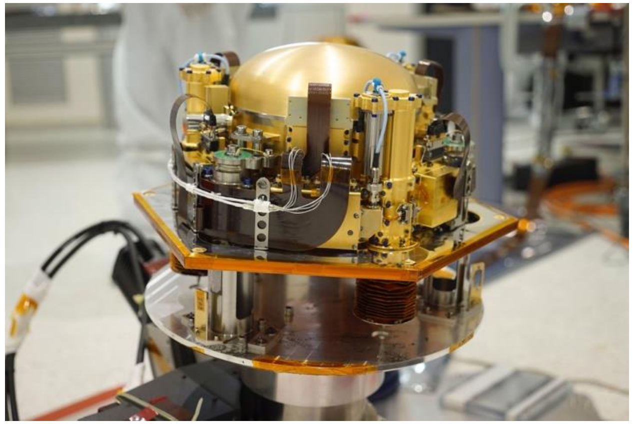
2. Machine learning in seismic noise application

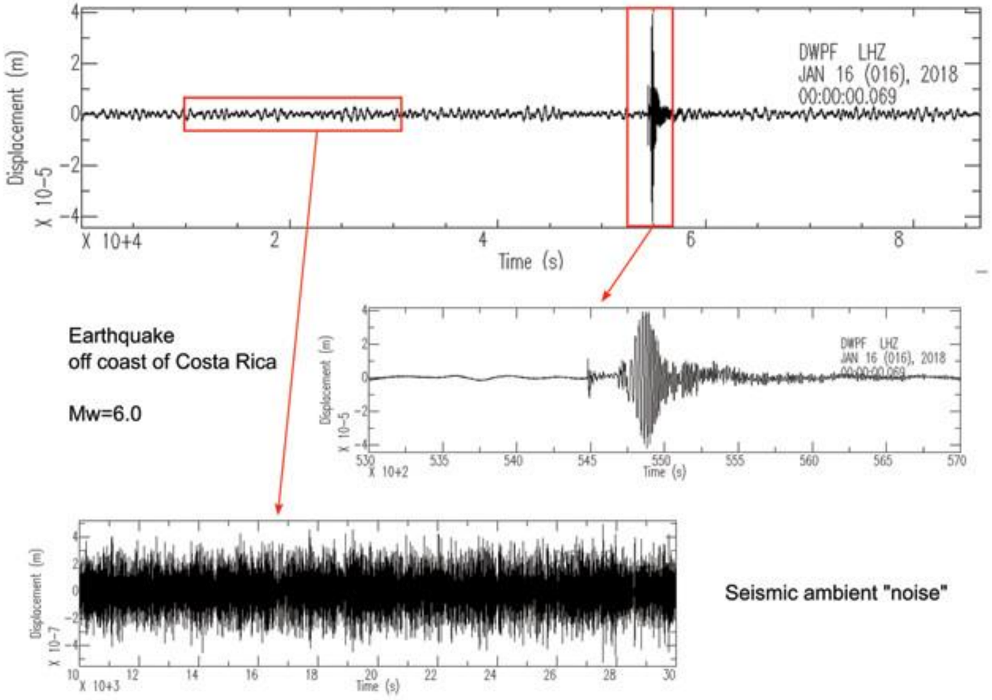
3. Traditional denoising methods

4. Machine learning denoising

1. Introduction

Definition and sources of seismic noise





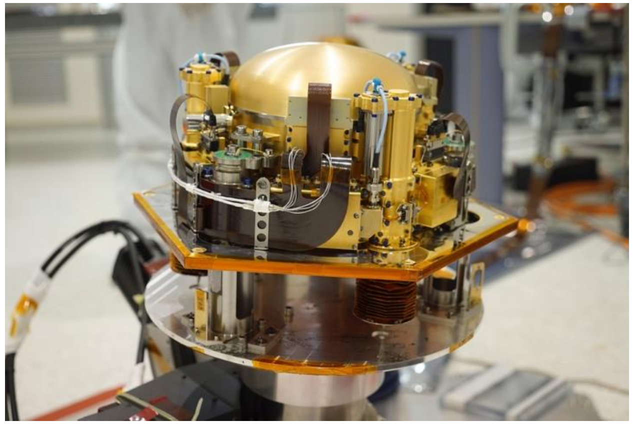
Definition: Seismic noise is interference from non-seismic signals that often masks useful seismic events (earthquakes, landslide, etc).

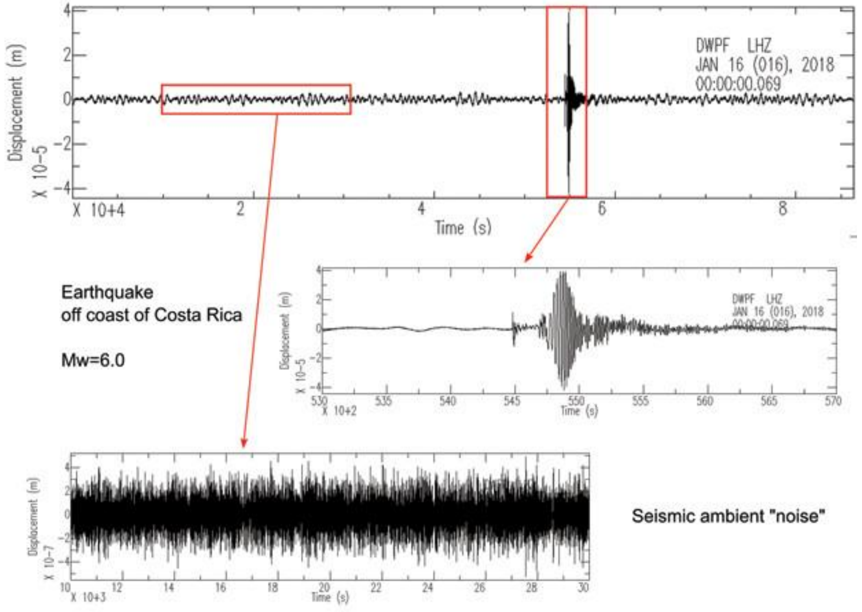
Sources: Natural noise (ocean waves, wind, volcanic activity) and anthropogenic noise (traffic, industrial activity, urban environments).

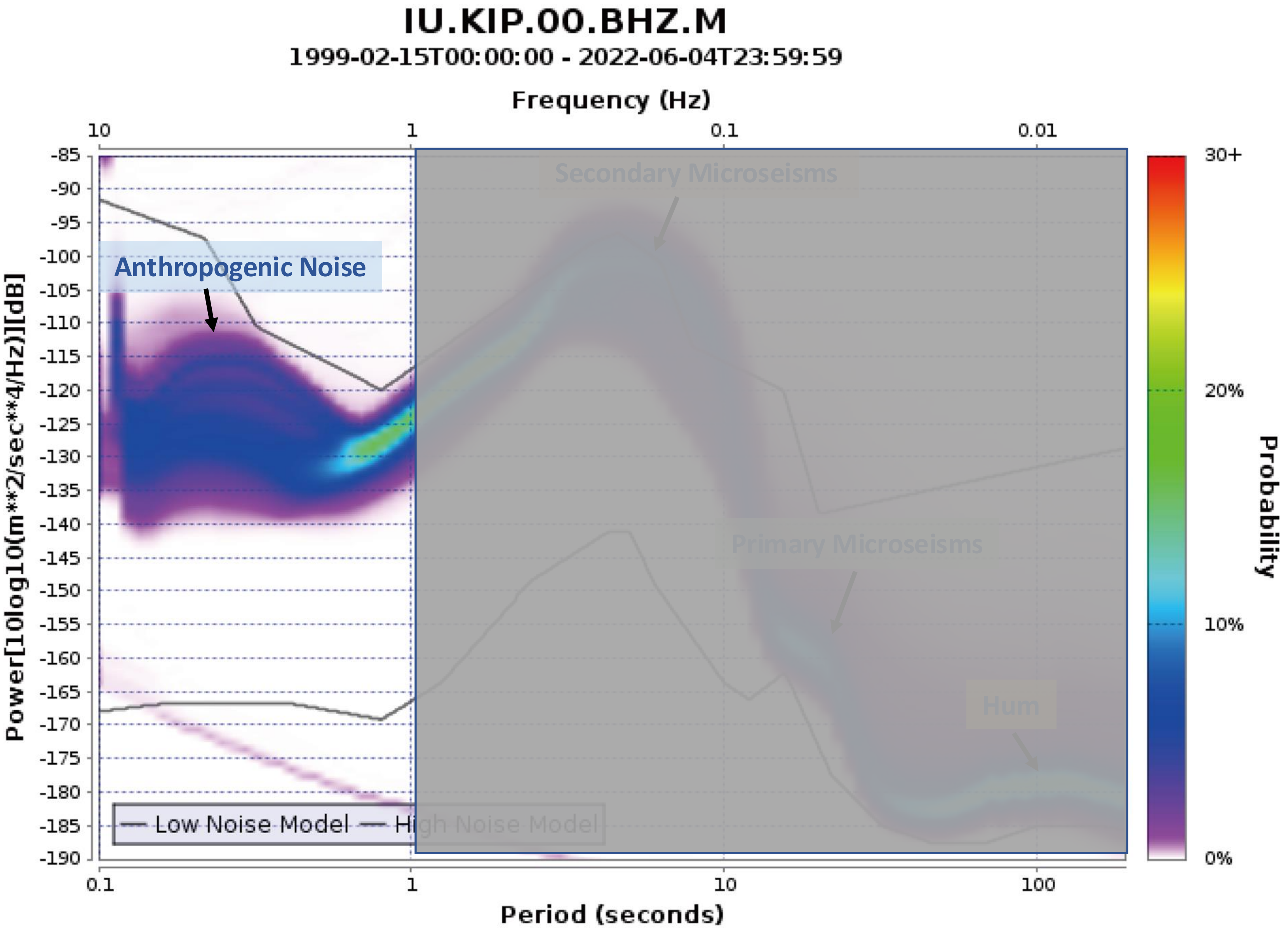
Impact on seismology: Reduces the signal-to-noise ratio, affecting the detection and imaging quality of seismic events.

1. Introduction

Seismic noise

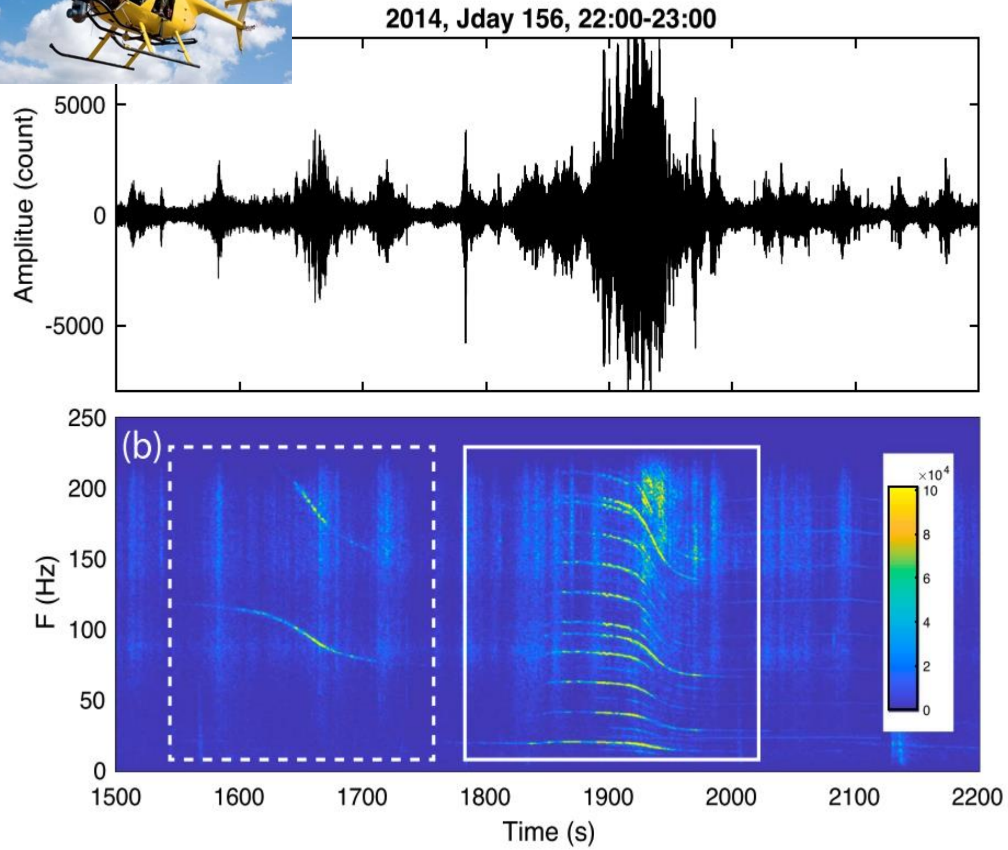




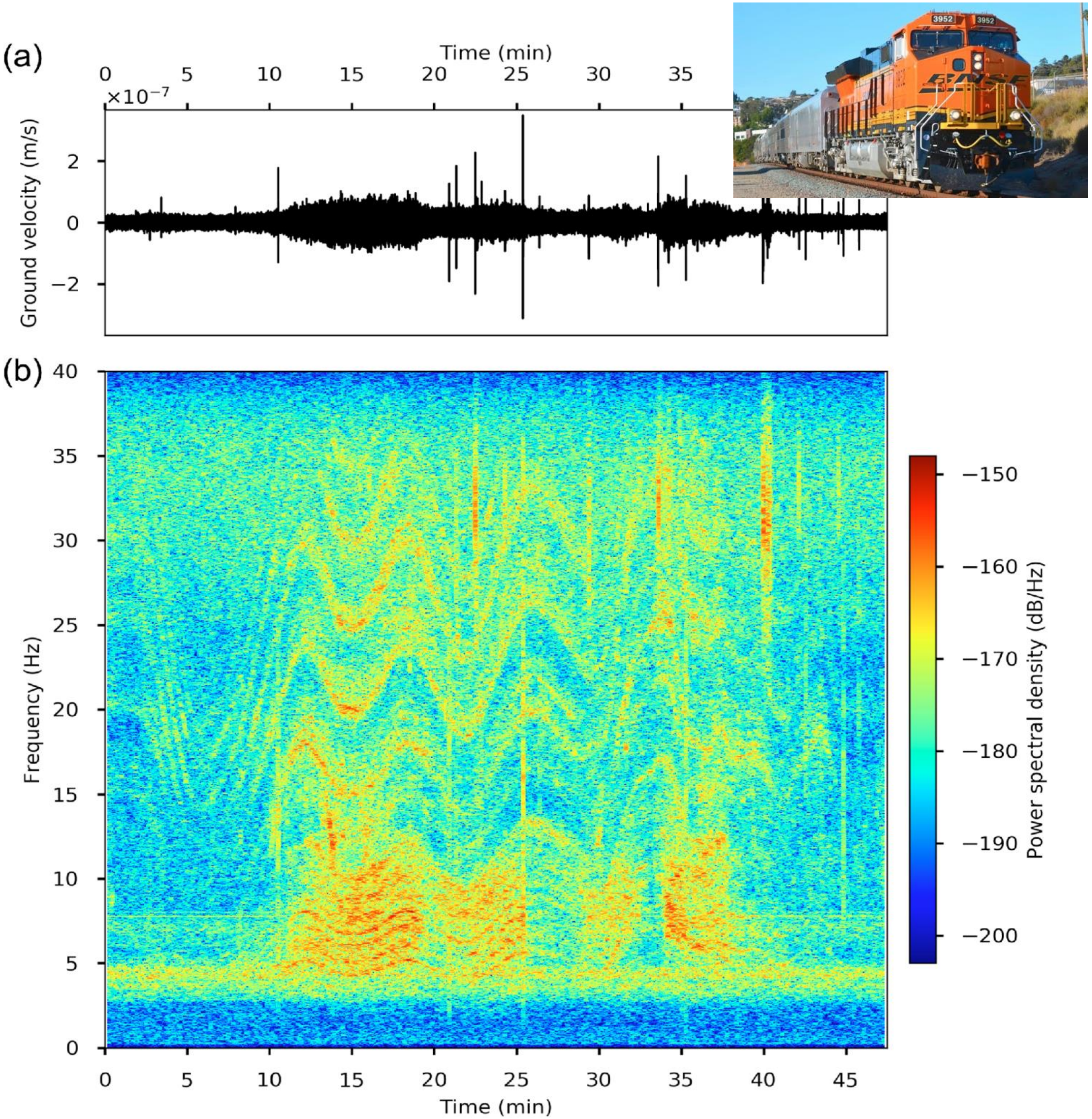


1. Introduction

Anthropogenic noise (> 1 Hz)



- Frequency band: 10-220 Hz

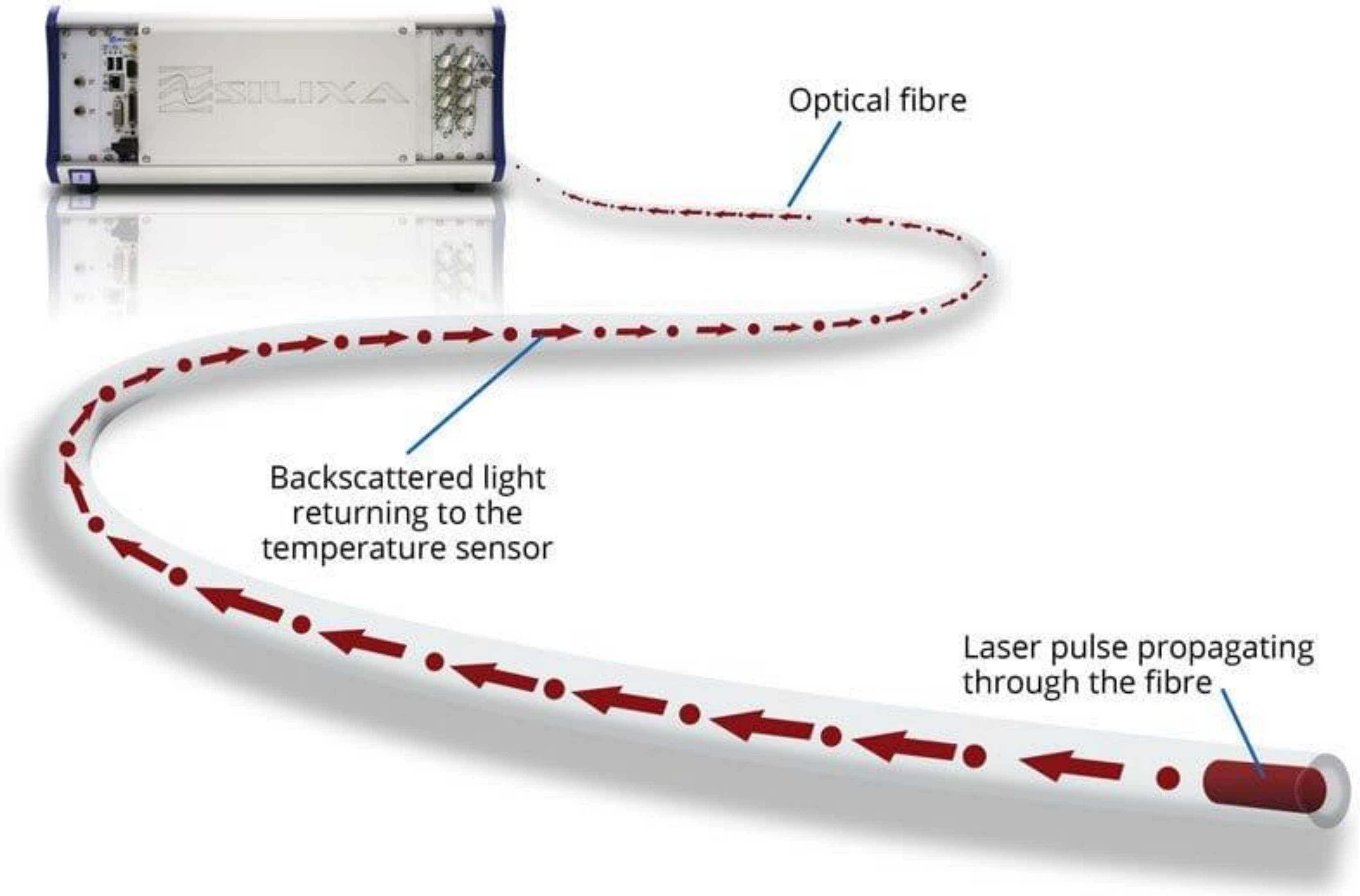


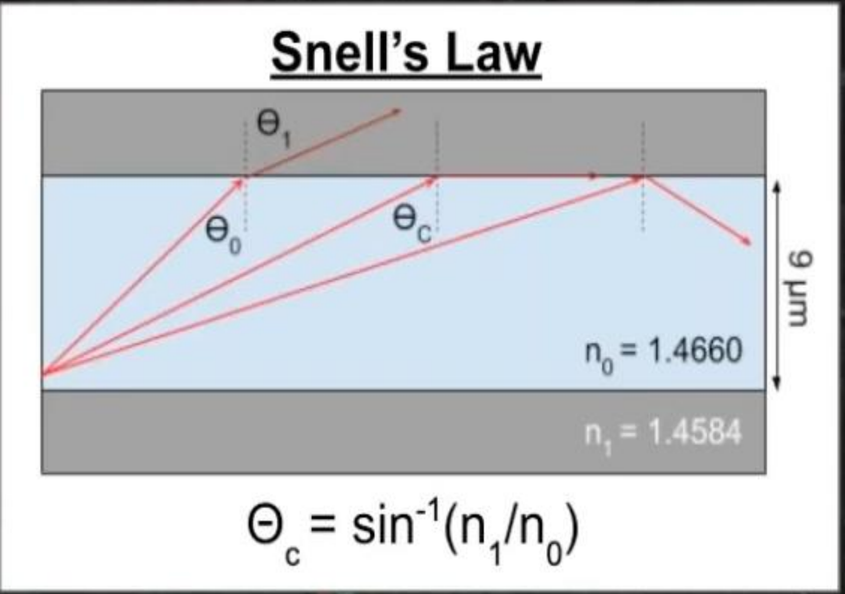
- Frequency band: 5-40 Hz

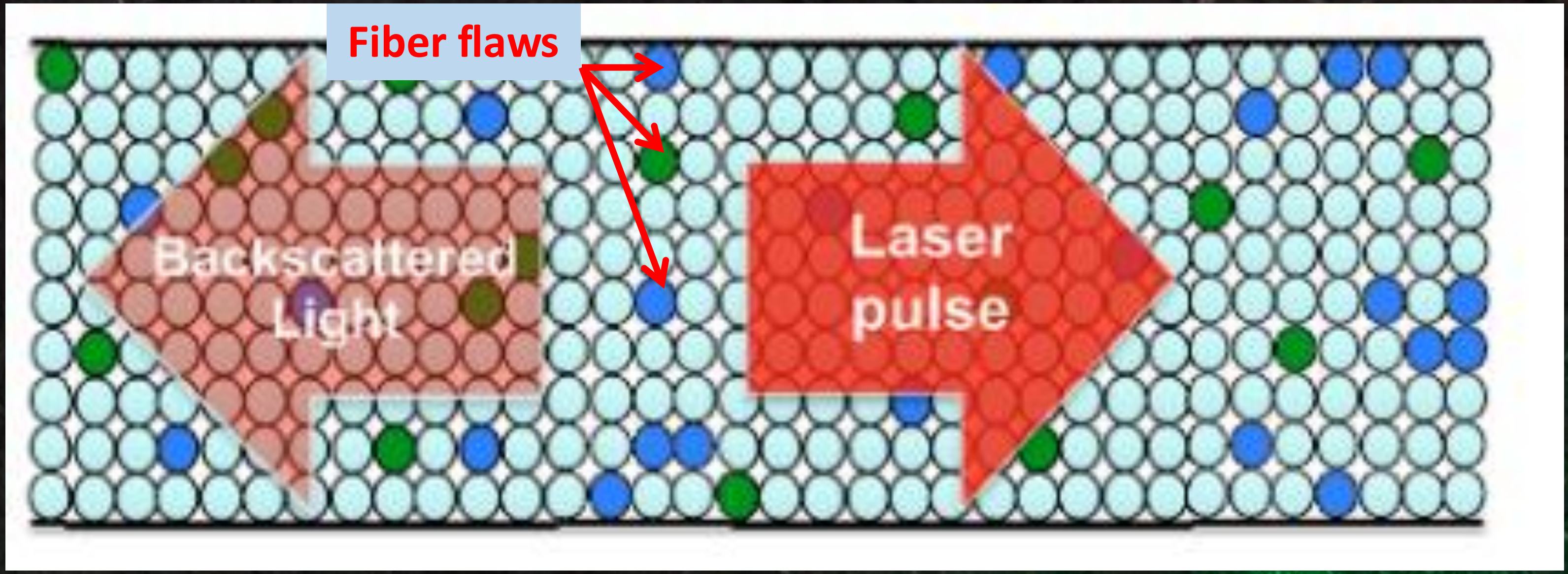
(Pinzon-Rincon et al., 2017)

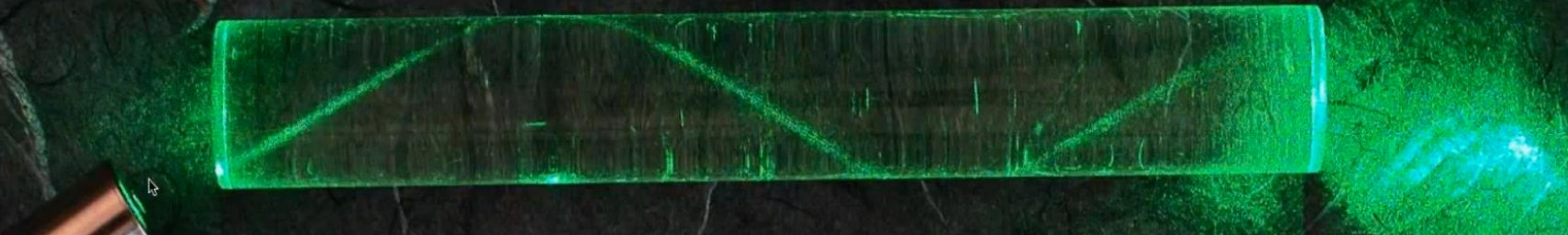
Chapter 4

DAS data









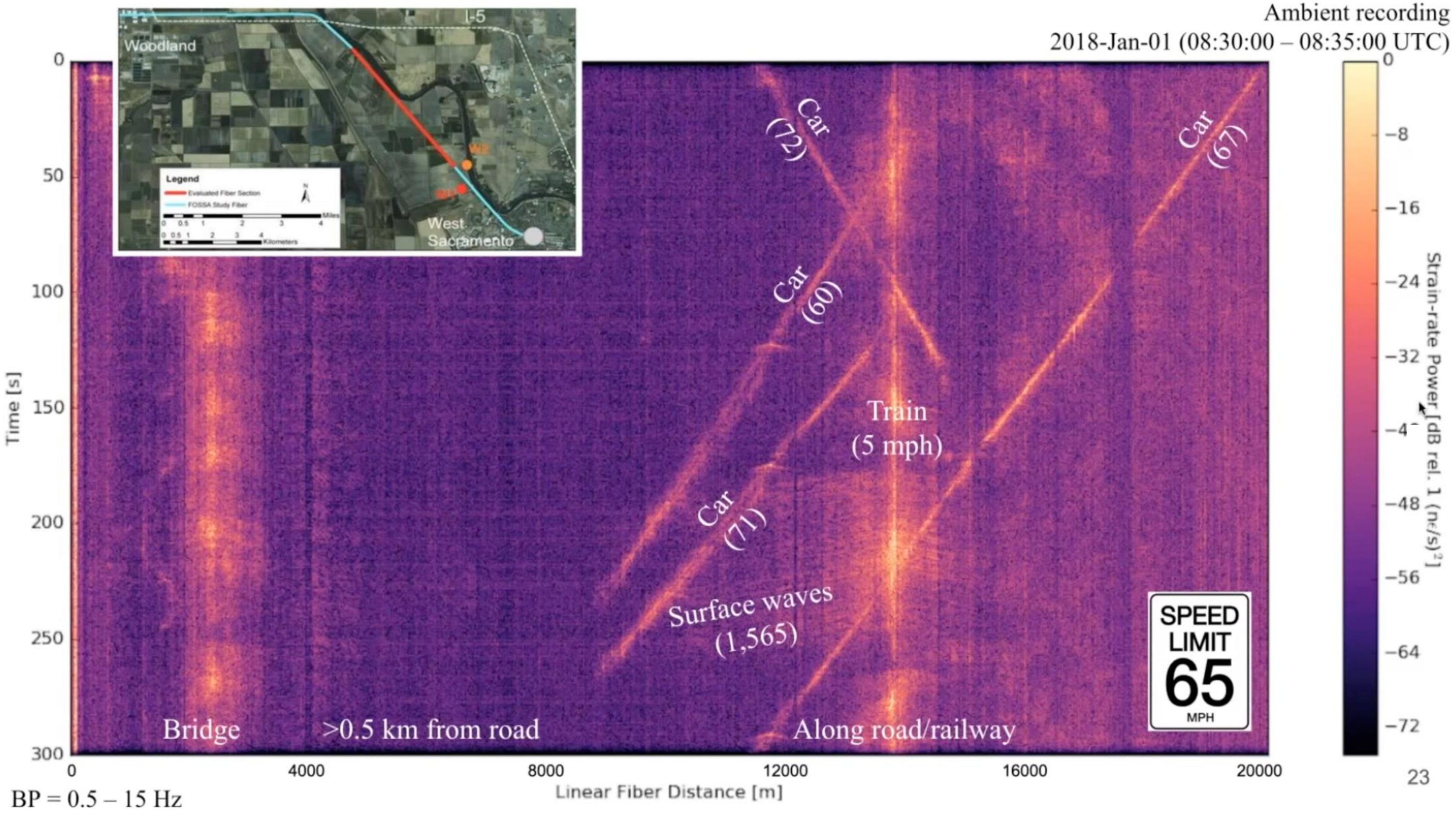
Number of photons = Laser energy / (hc / λ) * Pulse duration

Number of photons = 1 mW / (6.26e-34 Js * 3e8 m/s / 1550 nm) * 30 ns

Number of photons = 247,603,834 per pulse

1. Introduction

Anthropogenic noise (> 1 Hz)



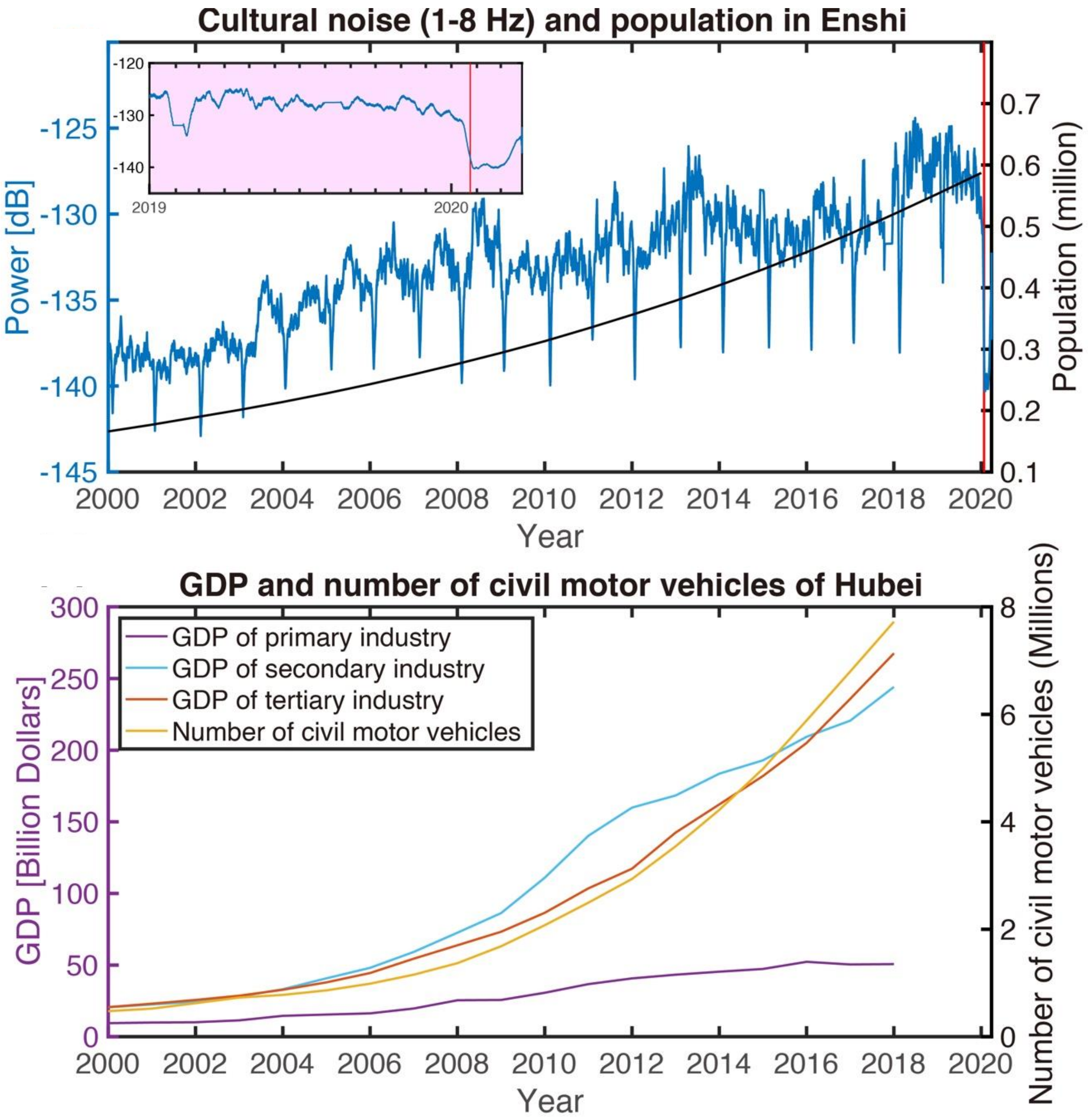
(Lindsey et al., 2020)

1. Introduction

Anthropogenic noise (> 1 Hz)

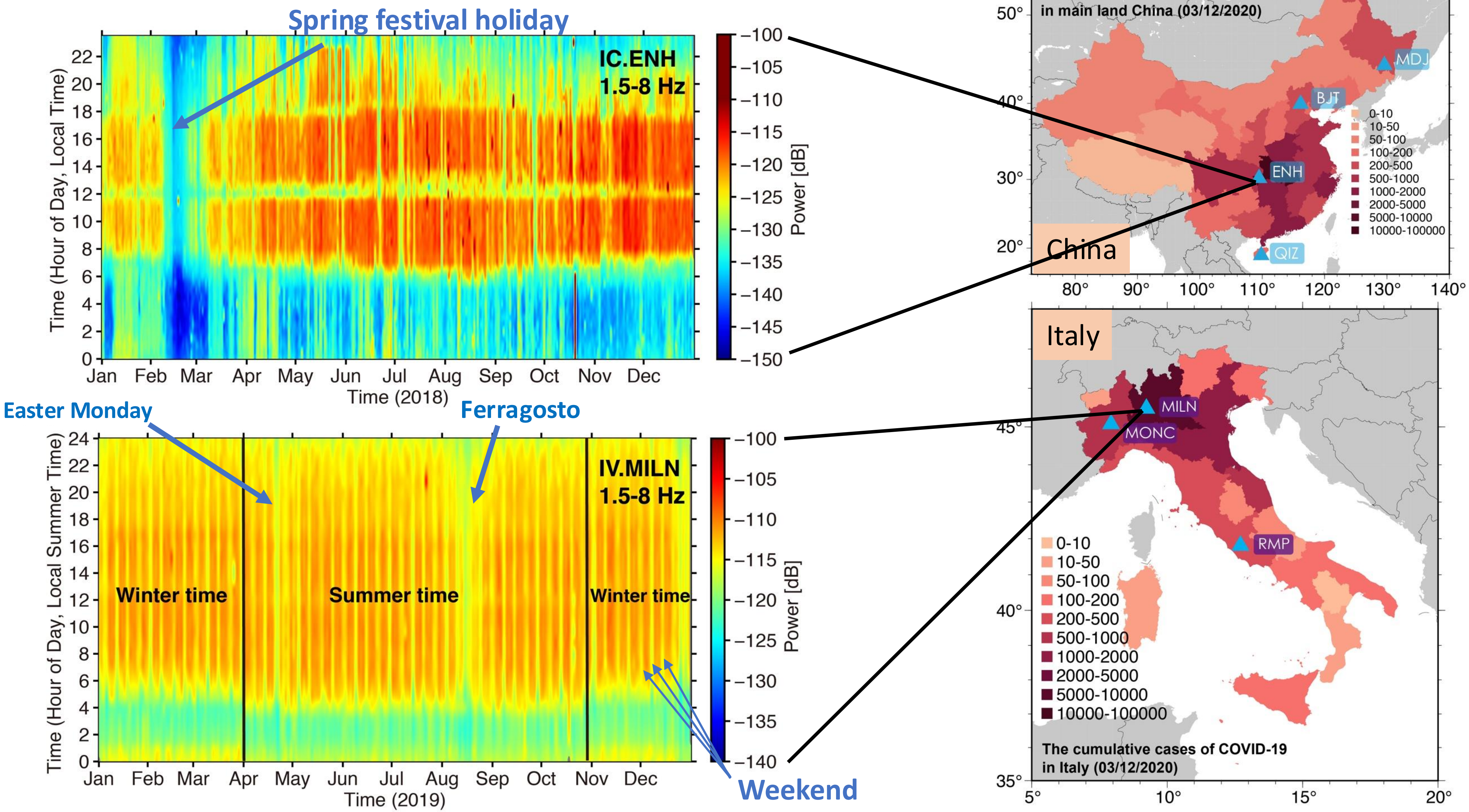
1) What can we do with anthropogenic noise?

1. Introduction



1. Introduction

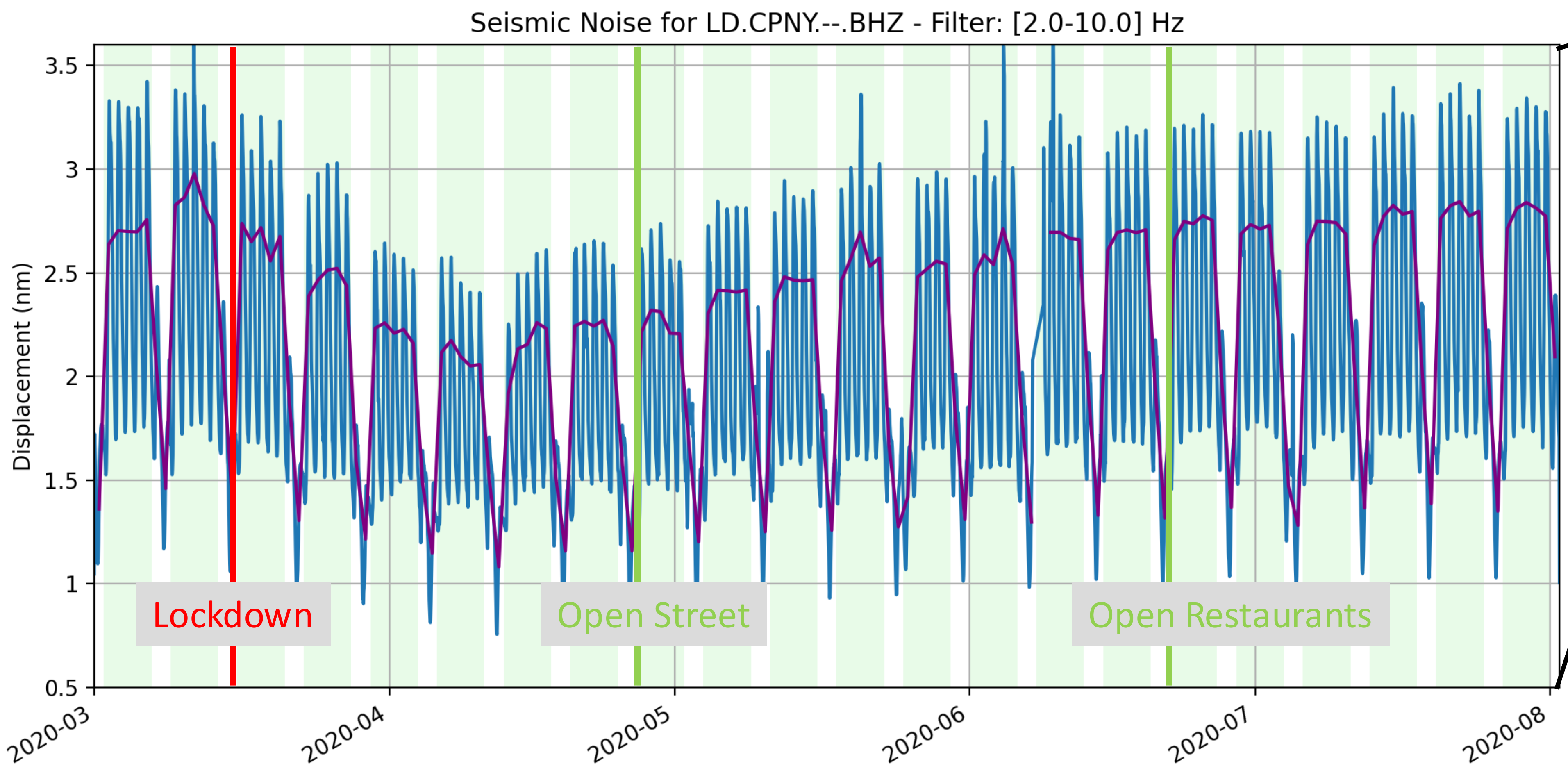
Seismic noise and living habits

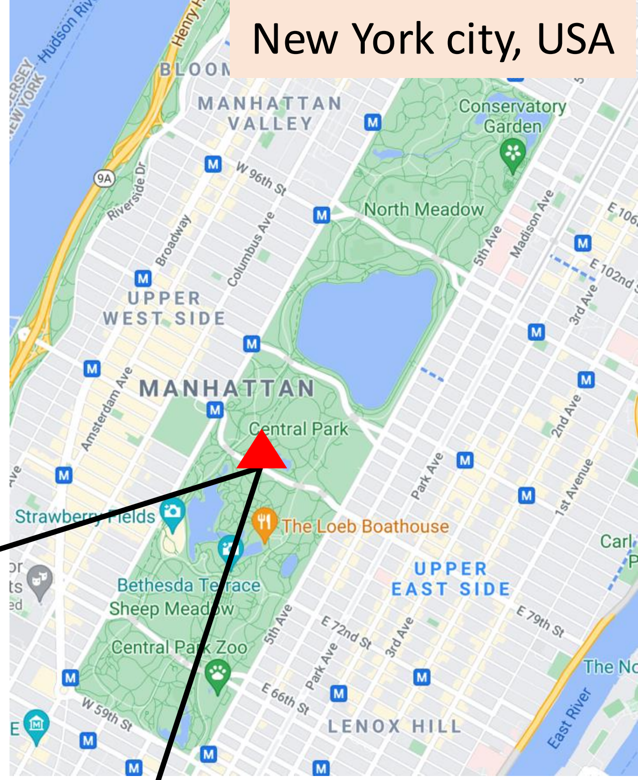


1. Introduction

COVID-19 lockdown effect on seismic noise,
New York, USA

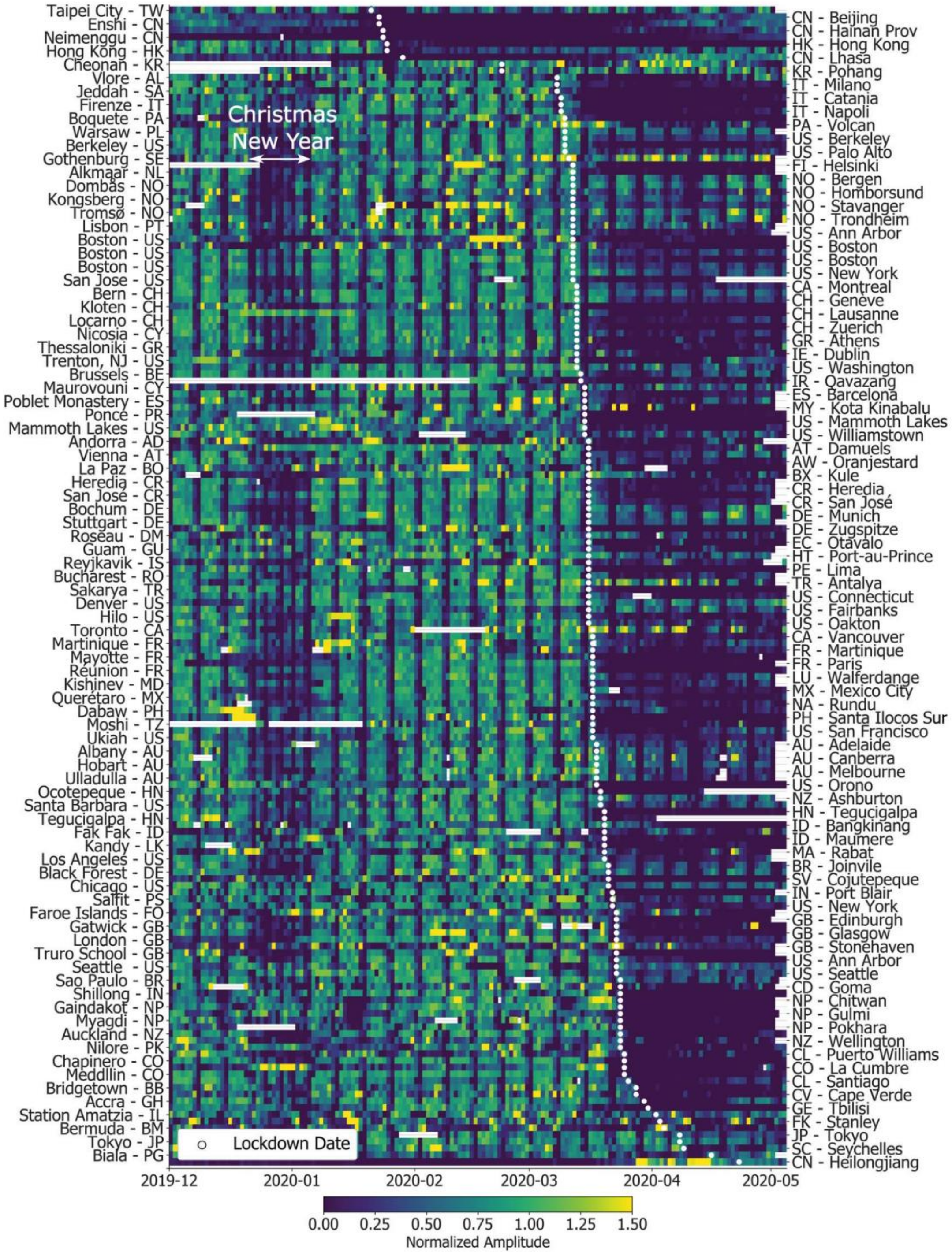
- **A maximum 25% noise reduction**





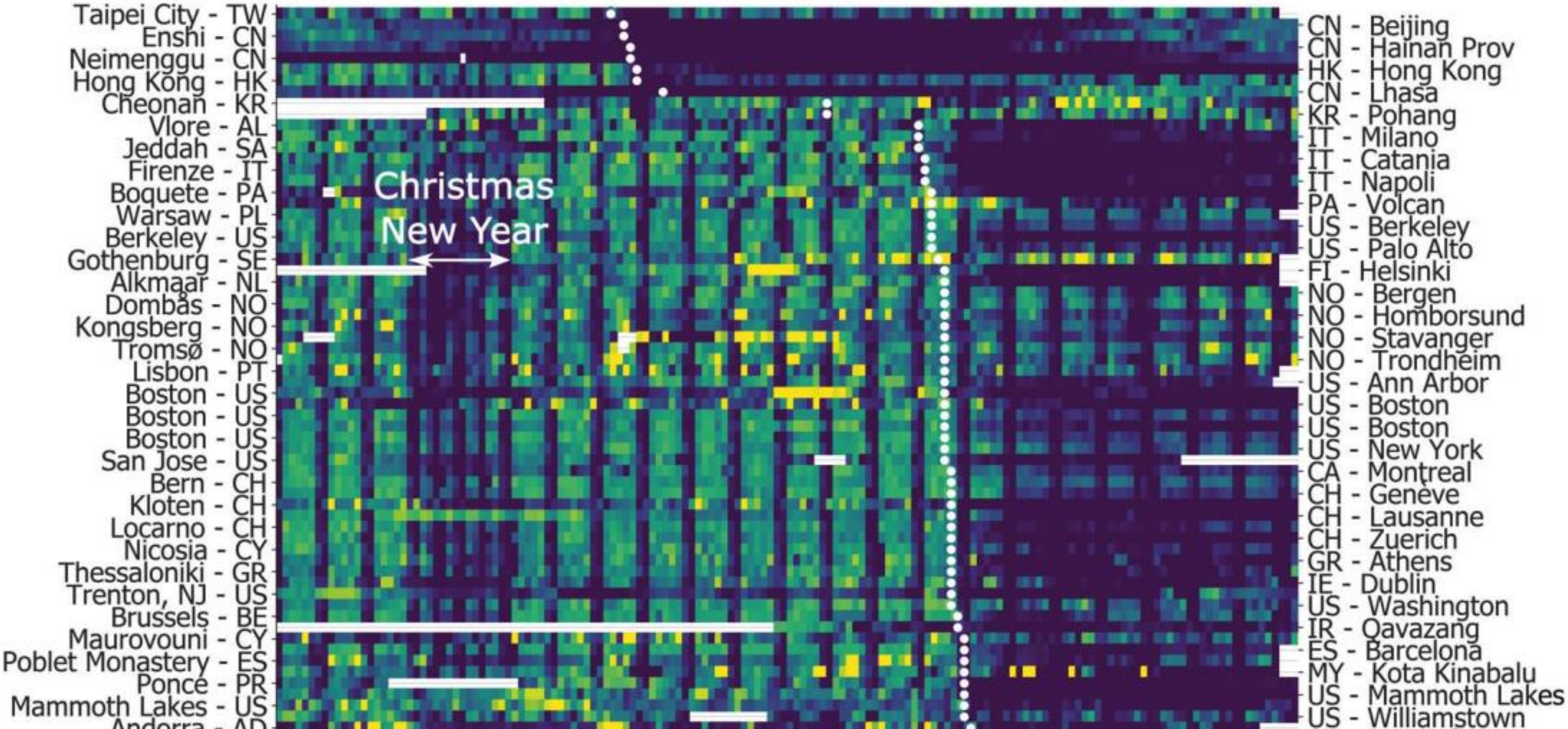
1. Introduction

Global quieting due to COVID-19



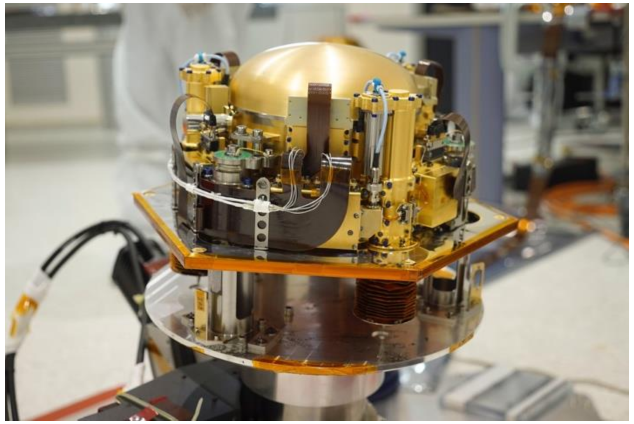
1. Introduction

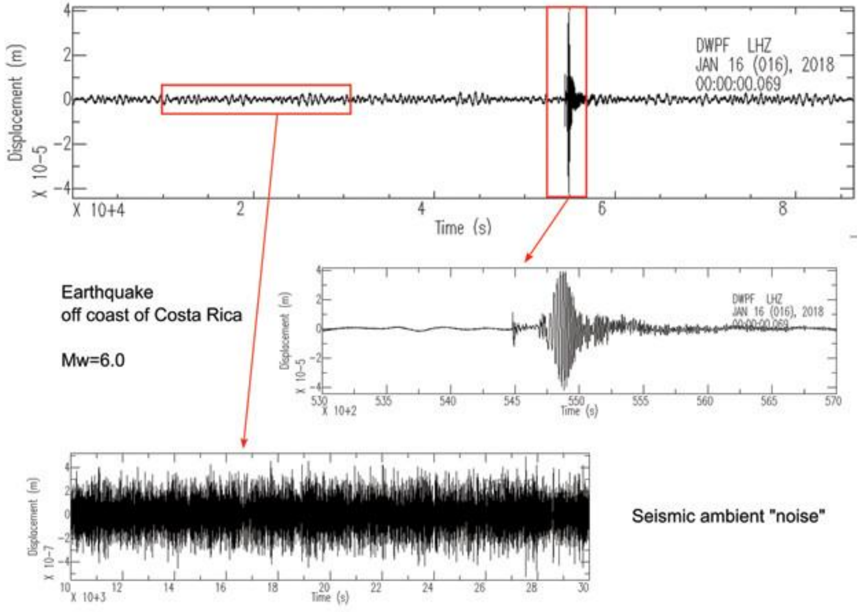
Global quieting due to COVID-19

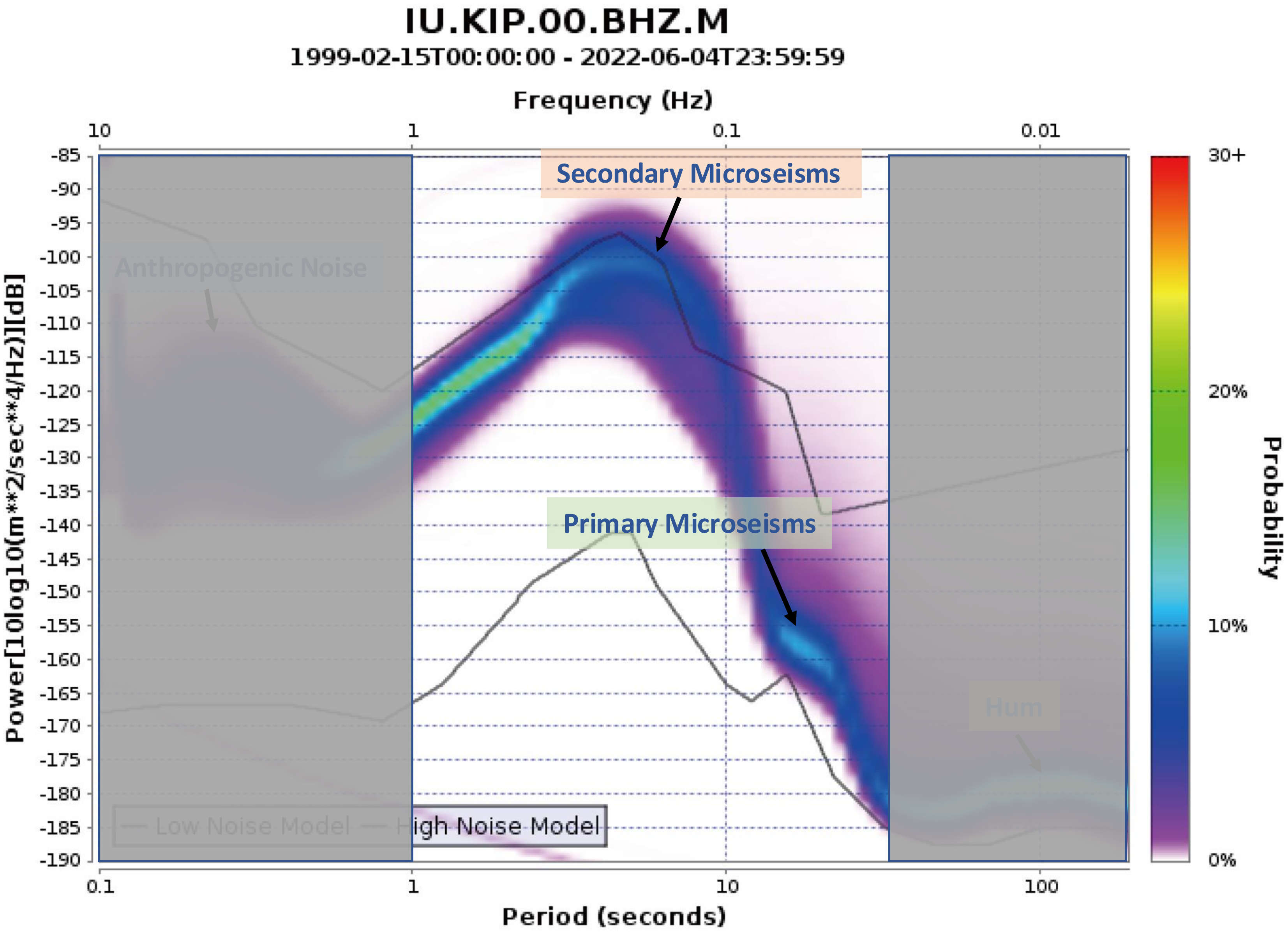


1. Introduction

Seismic noise

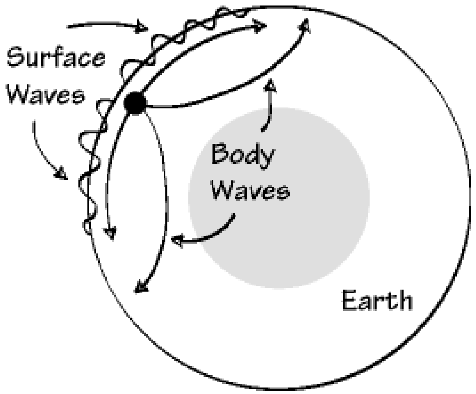






1. Introduction

The generation of microseisms



Seismic Wave

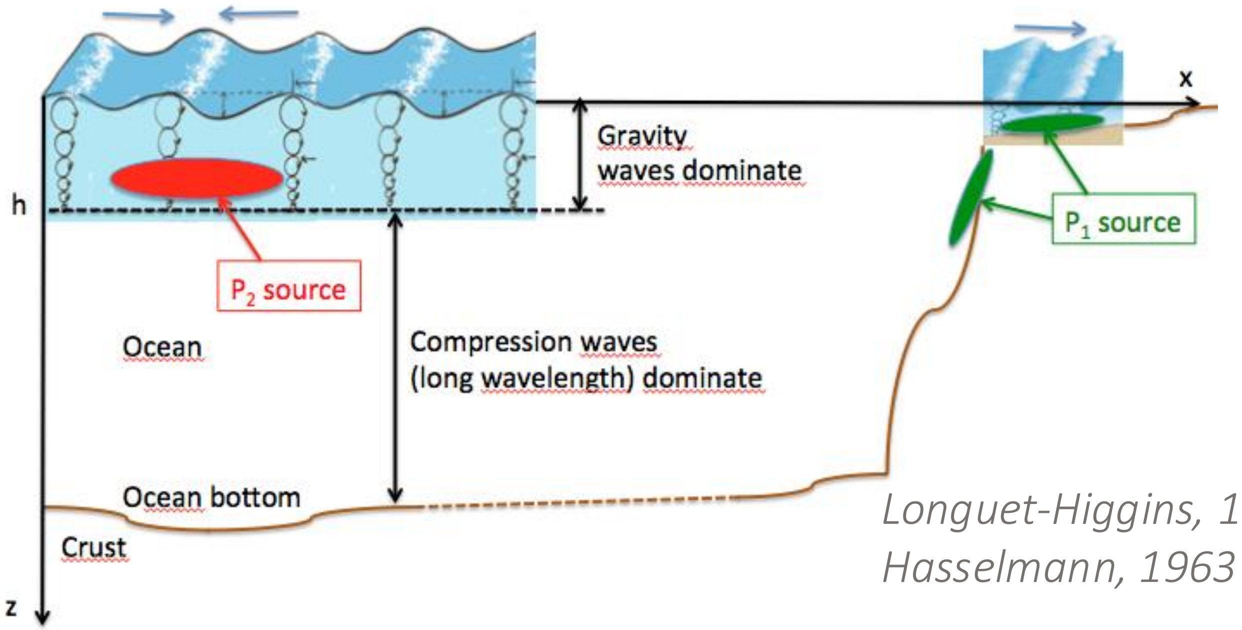
Surface Wave

Body Wave

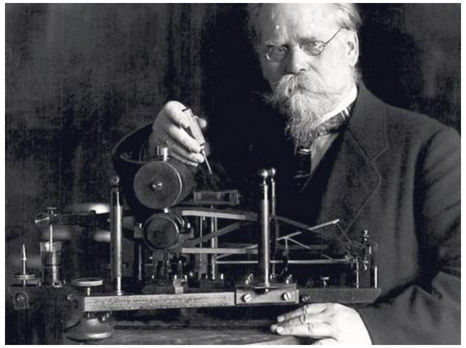
Mechanisms

Wind sea and swell → primary and secondary microseisms

Infragravity wave → hum



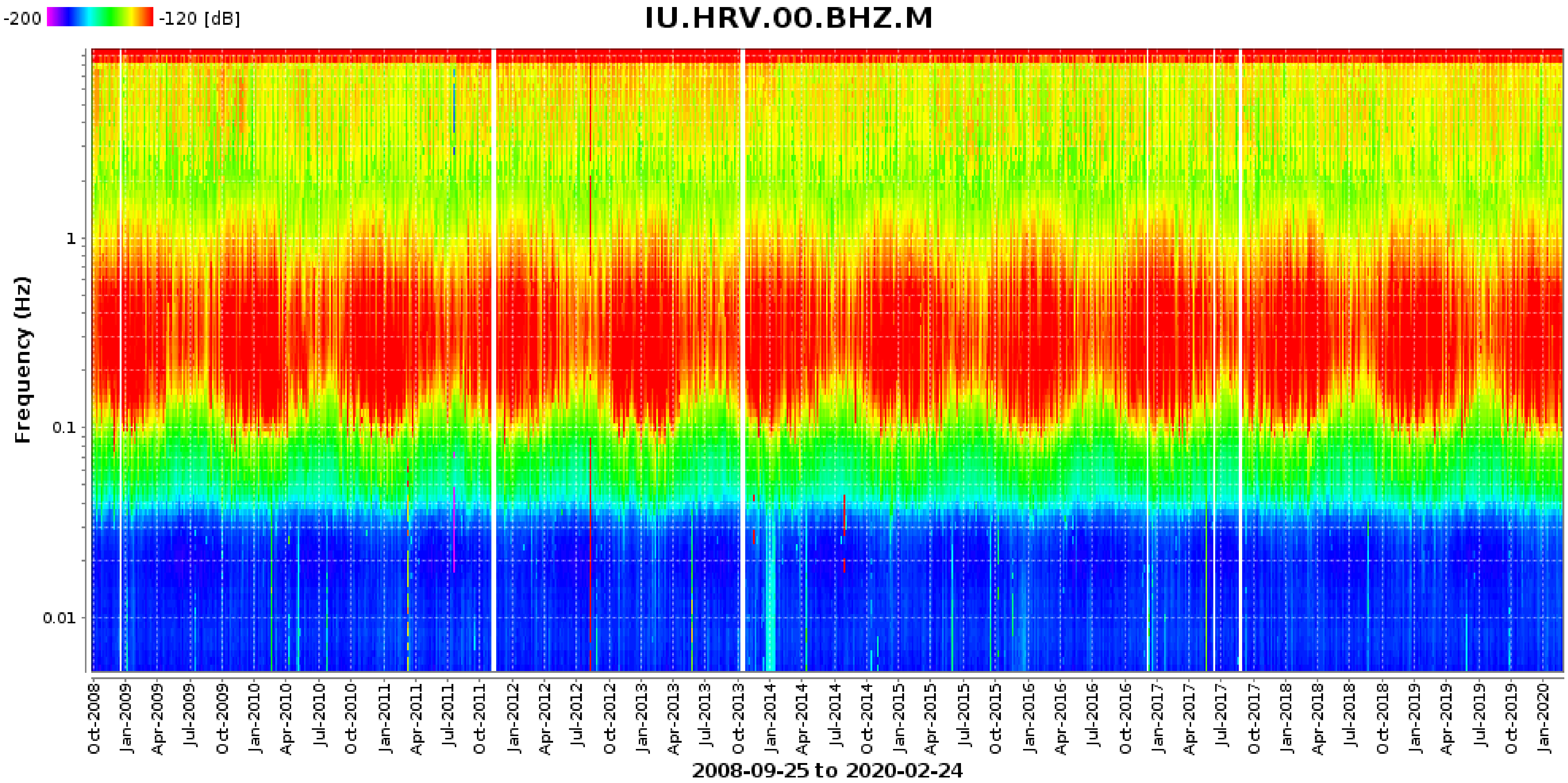
Longuet-Higgins, 1950
Hasselmann, 1963



Wiechert, 1904

1. Introduction

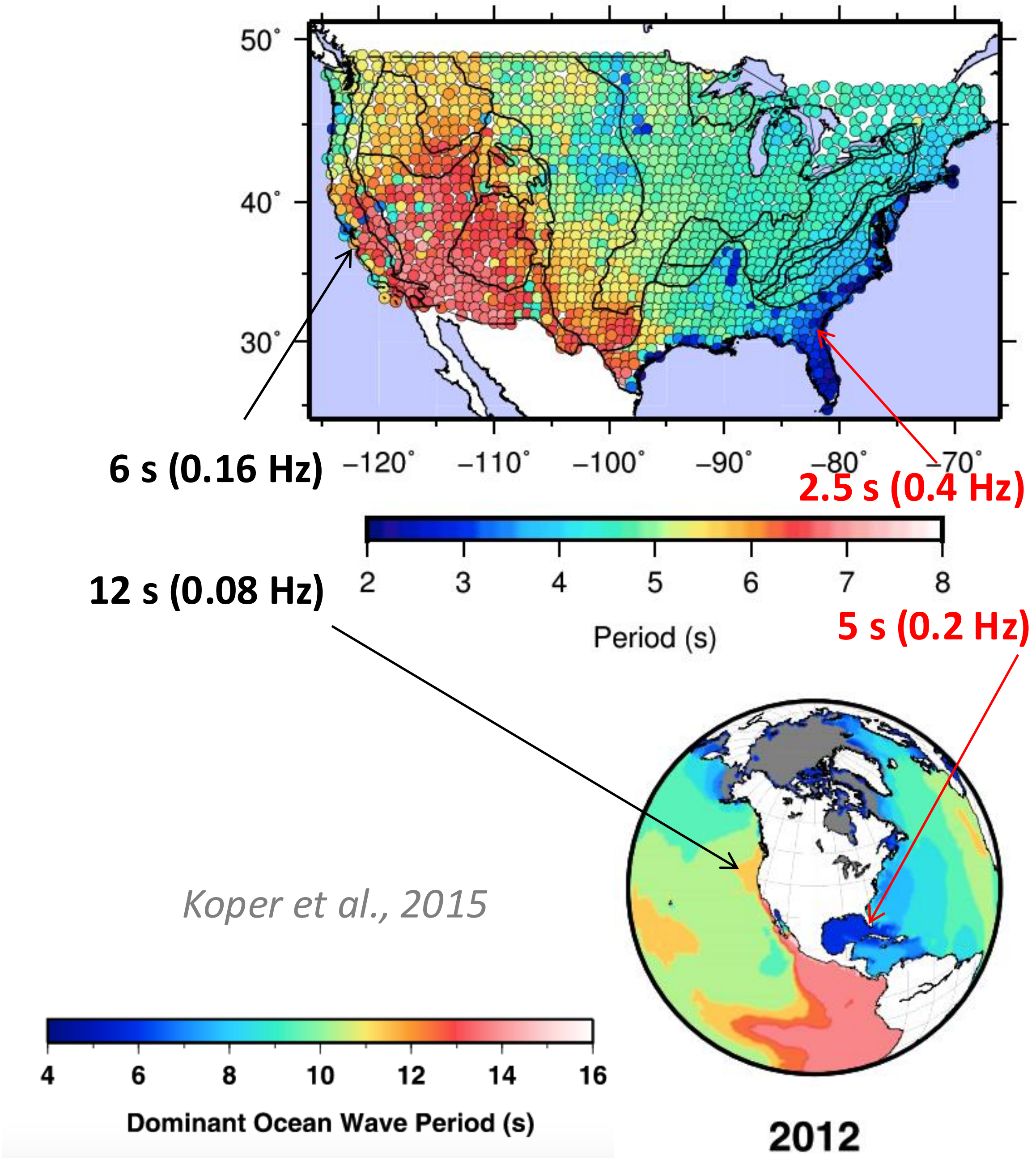
The variation of microseisms



1. Introduction

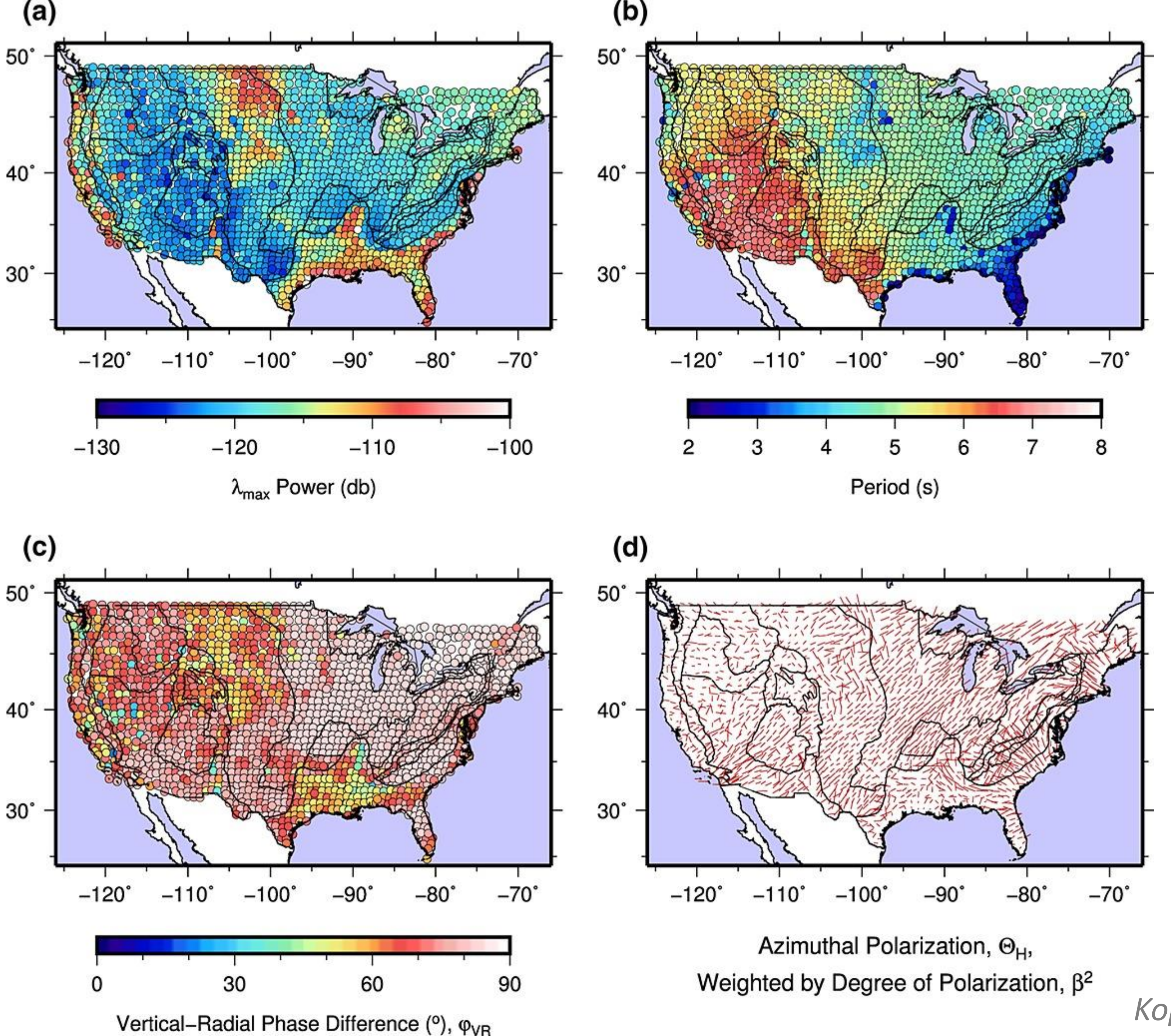
Secondary microseisms

The secondary frequency microseism propagate as half of the ocean period



Koper et al., 2015

1. Introduction

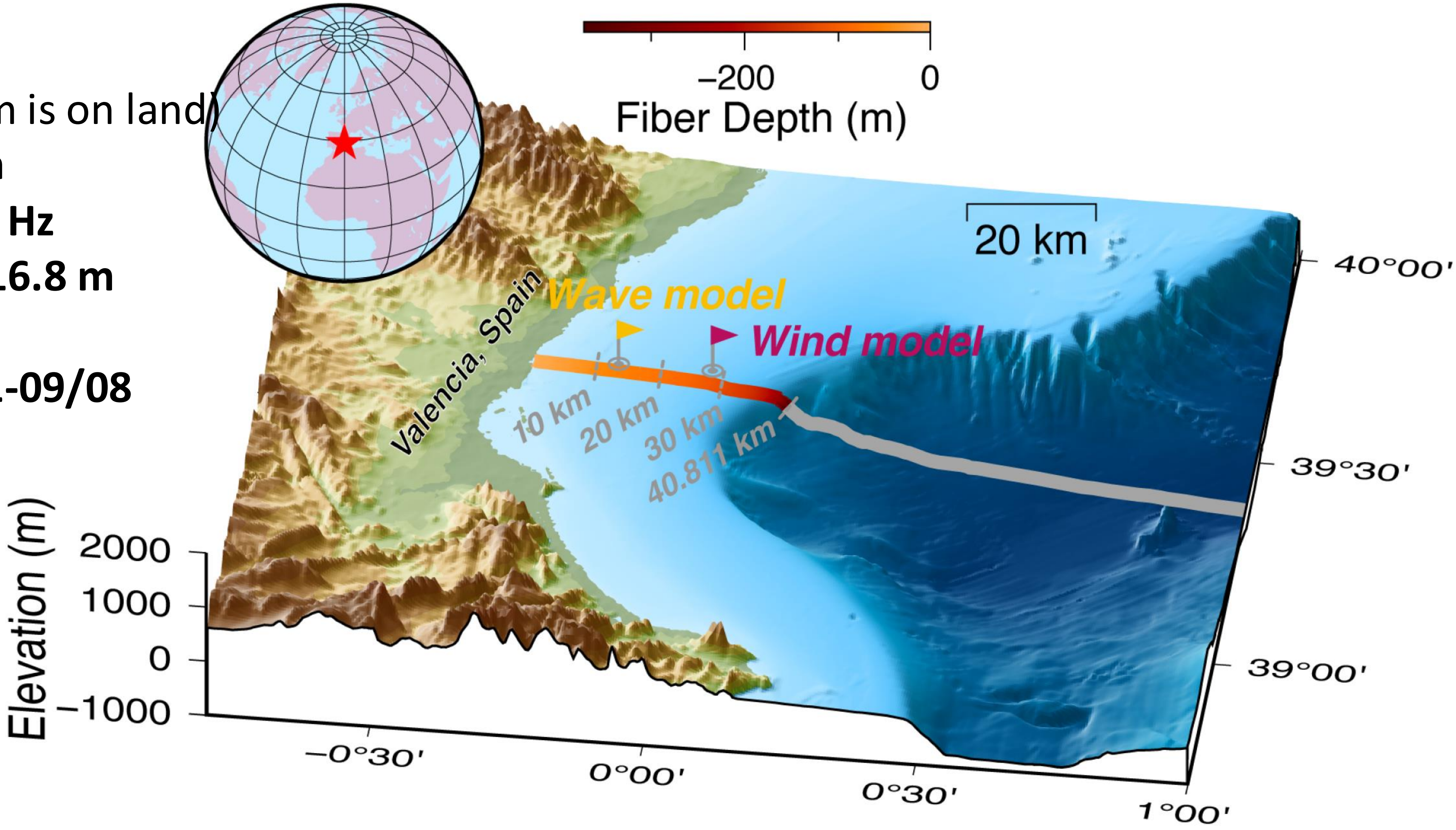


Koper et al., 2015

Chapter 4

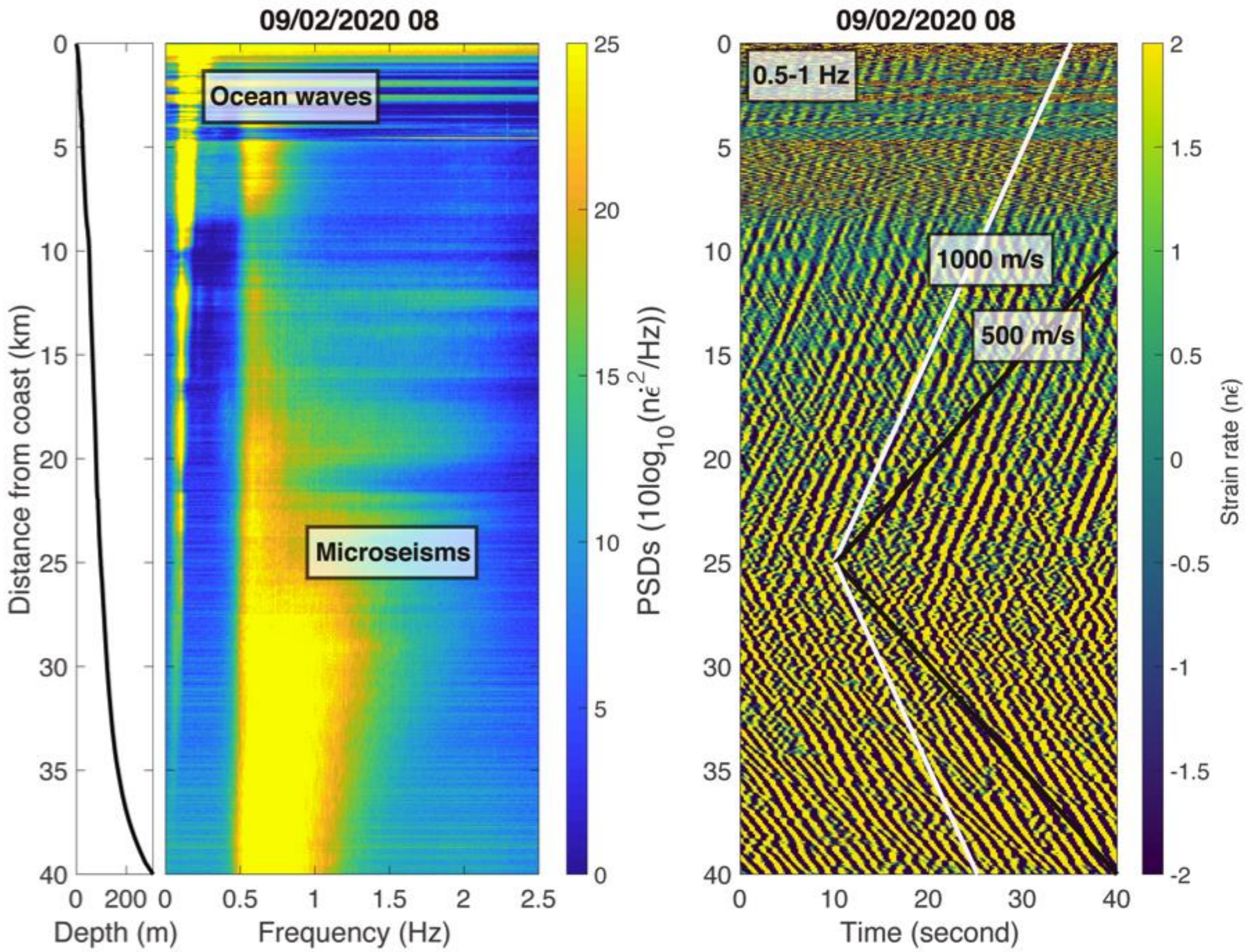
DAS data

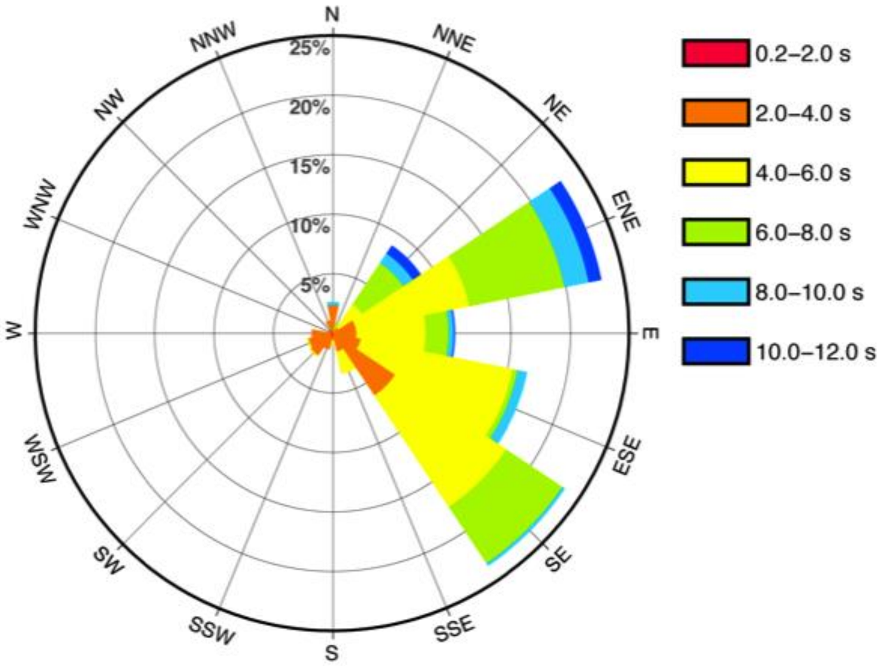
- **50 km** long (~9 km is on land)
- Burial depth **~1 m**
- Sample rate **1000 Hz**
- Channel spacing **16.8 m**
- **2977** channels
- **8** days data **09/01-09/08 2020**



Chapter 4

DAS data example

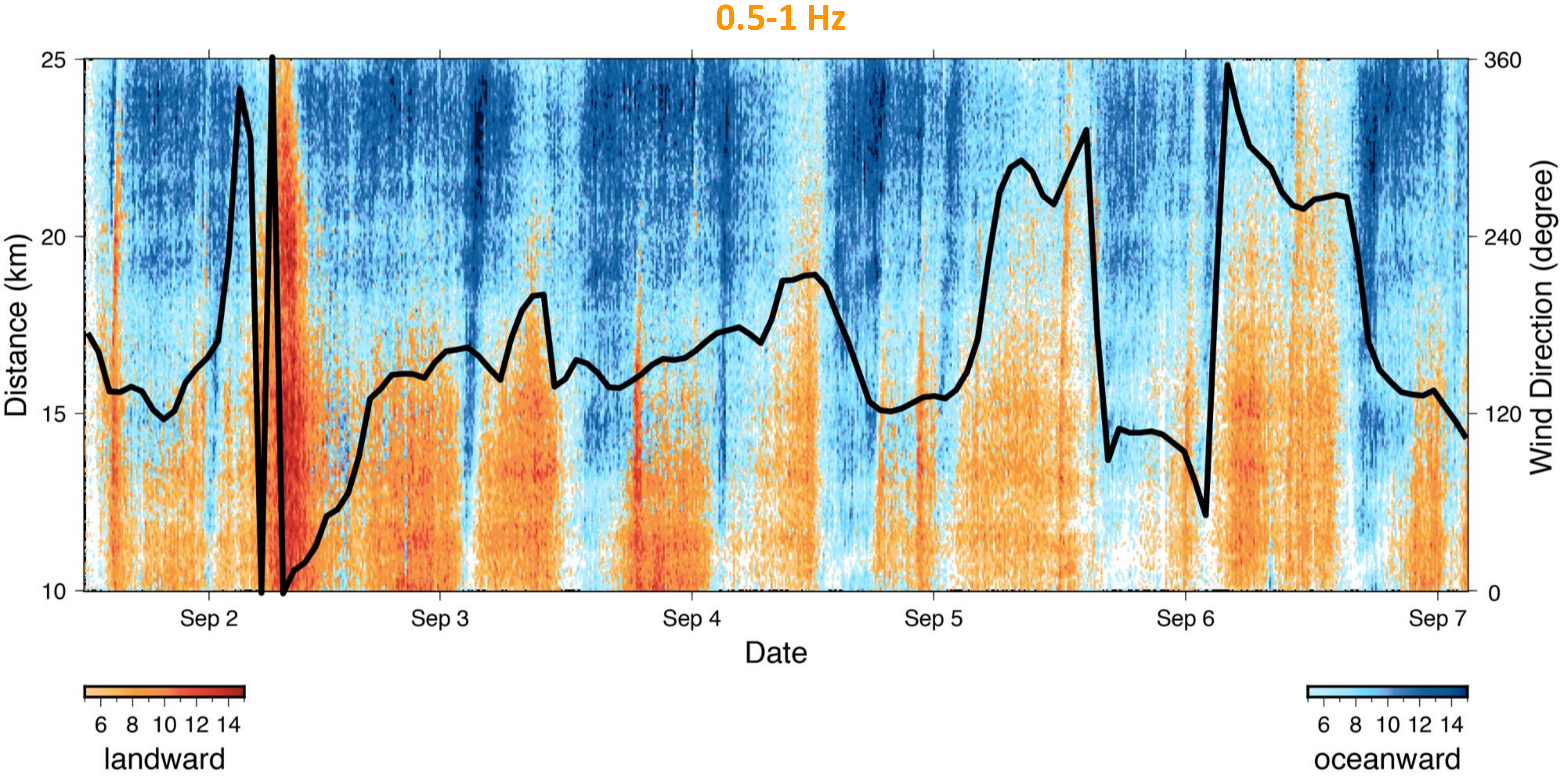




- The frequency of ocean waves propagating in the opposite direction is **0.25-0.5 Hz**

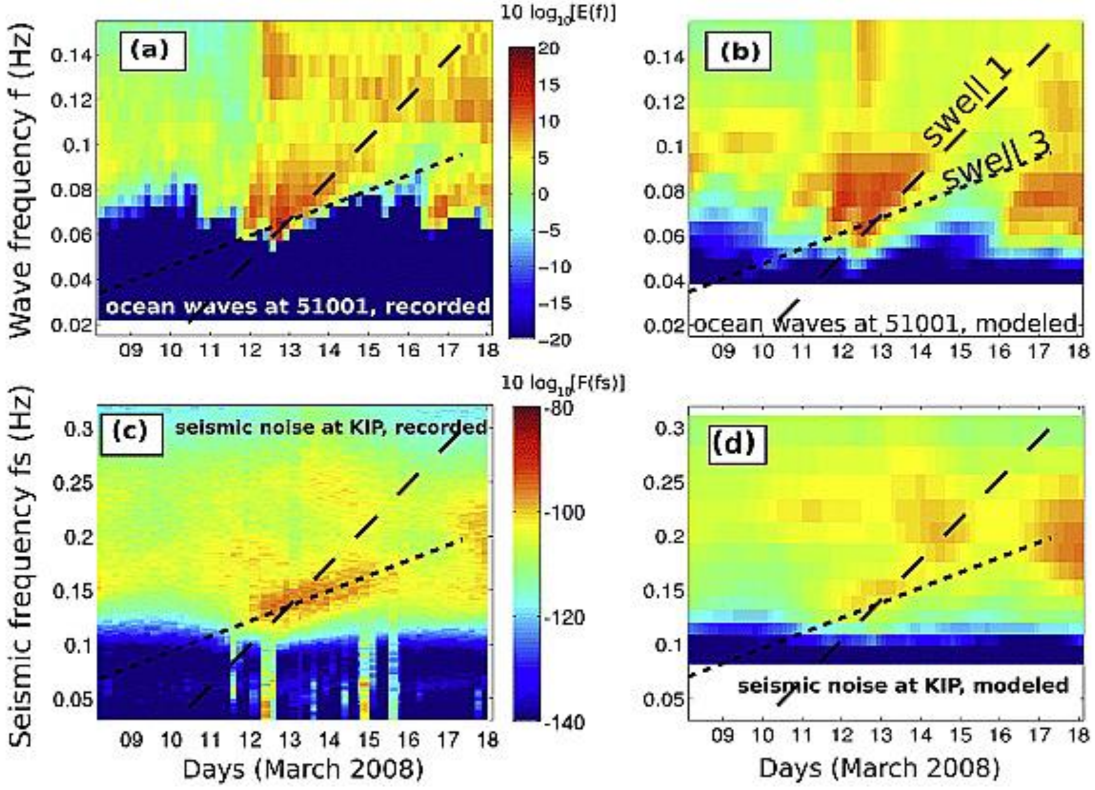
1. Introduction

Relationship with wind directions

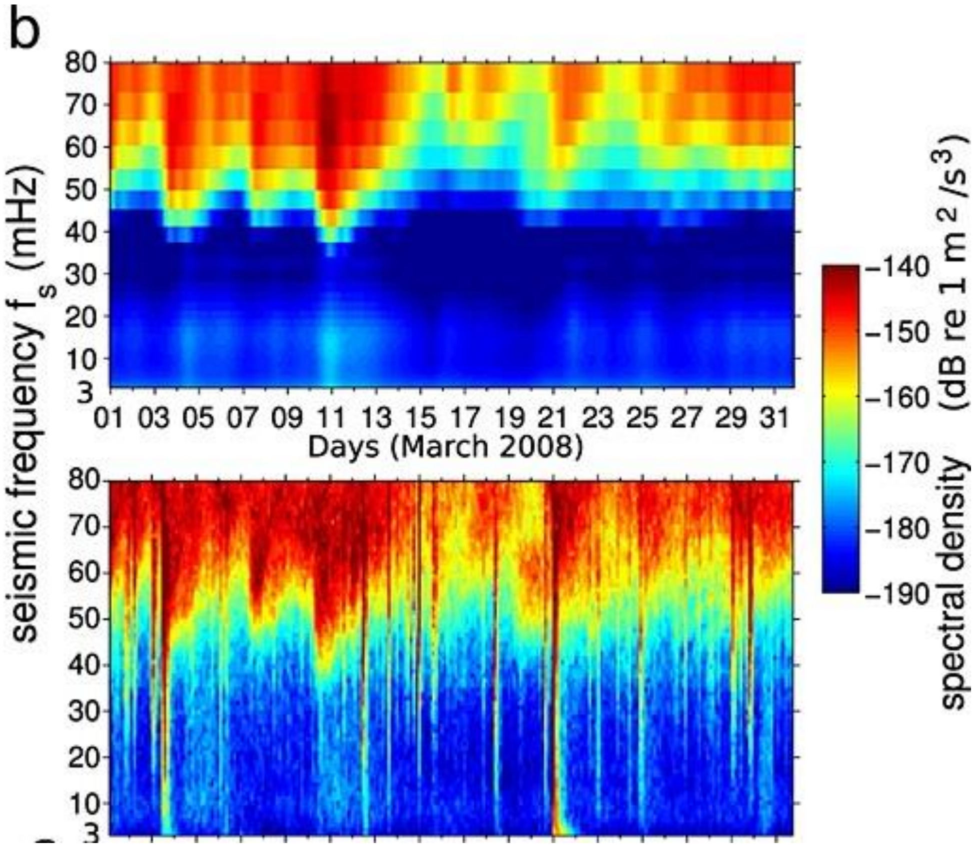


1. Introduction

Modeling of microseisms



Ardhuin et al., 2011



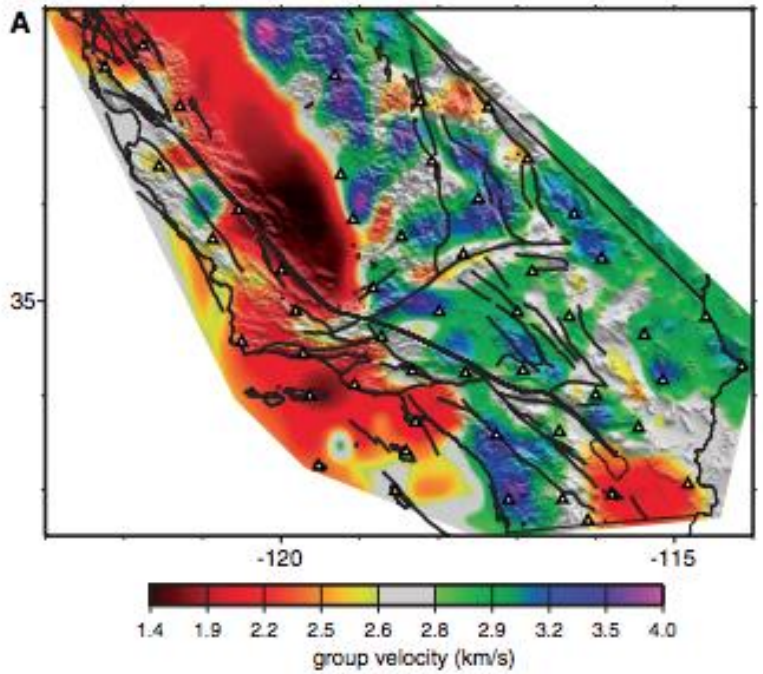
Ardhuin et al., 2015

1. Introduction

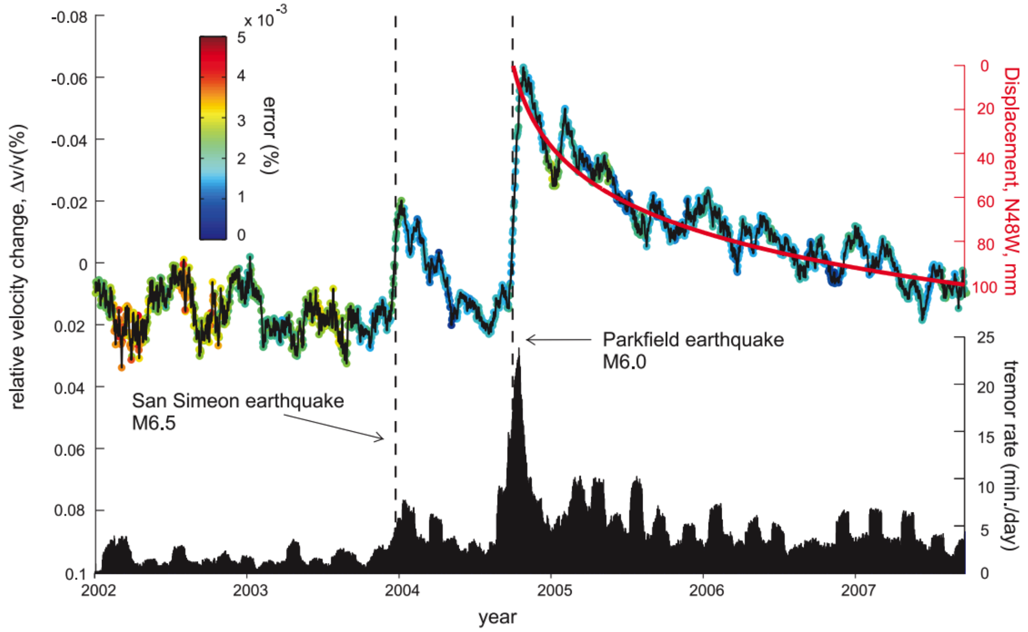
Uses of microseisms

Surface wave

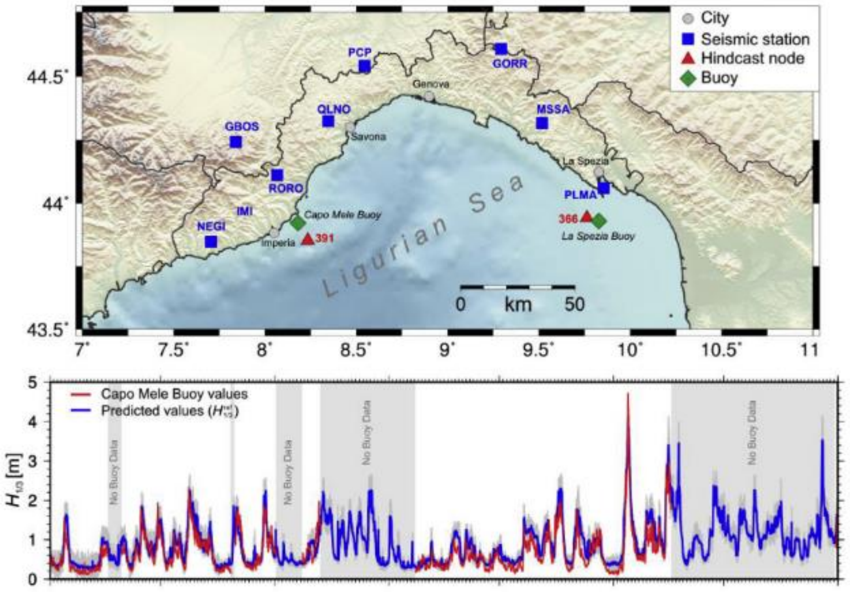
- Ambient noise tomography
- Monitor the underground structure
- Monitor ocean wave height near the coast



Shapiro et al., Science 2005



Brenguier et al., Science 2008



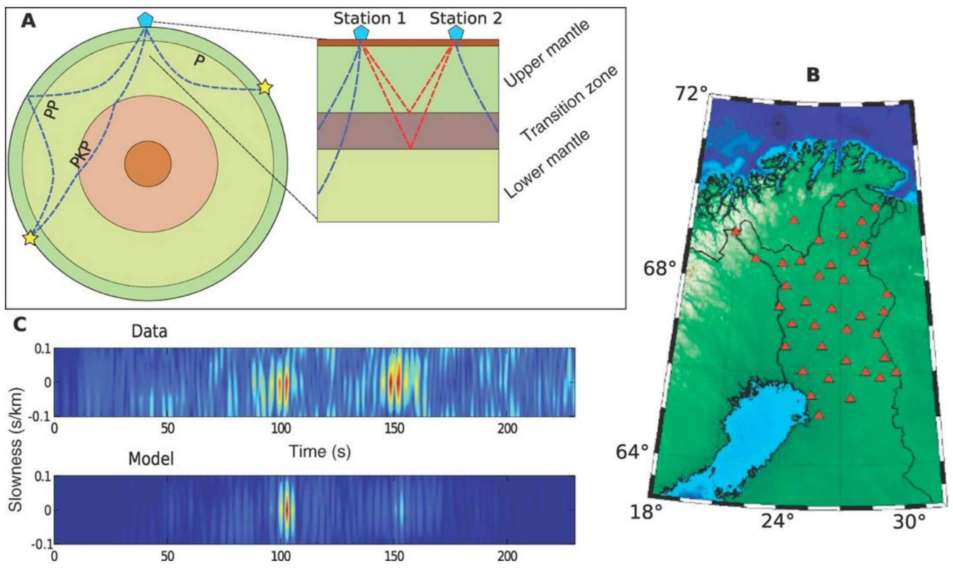
Ferretti et al., 2018

1. Introduction

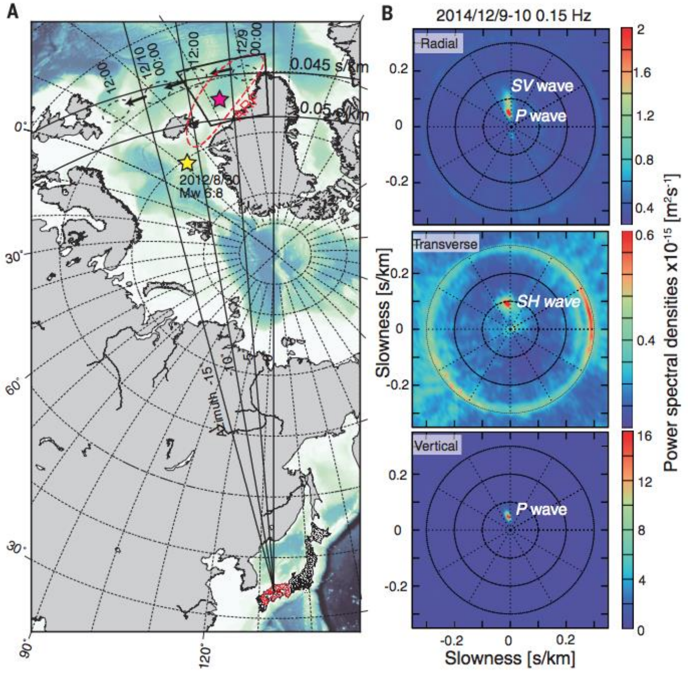
Uses of microseisms

Body wave (mostly P wave)

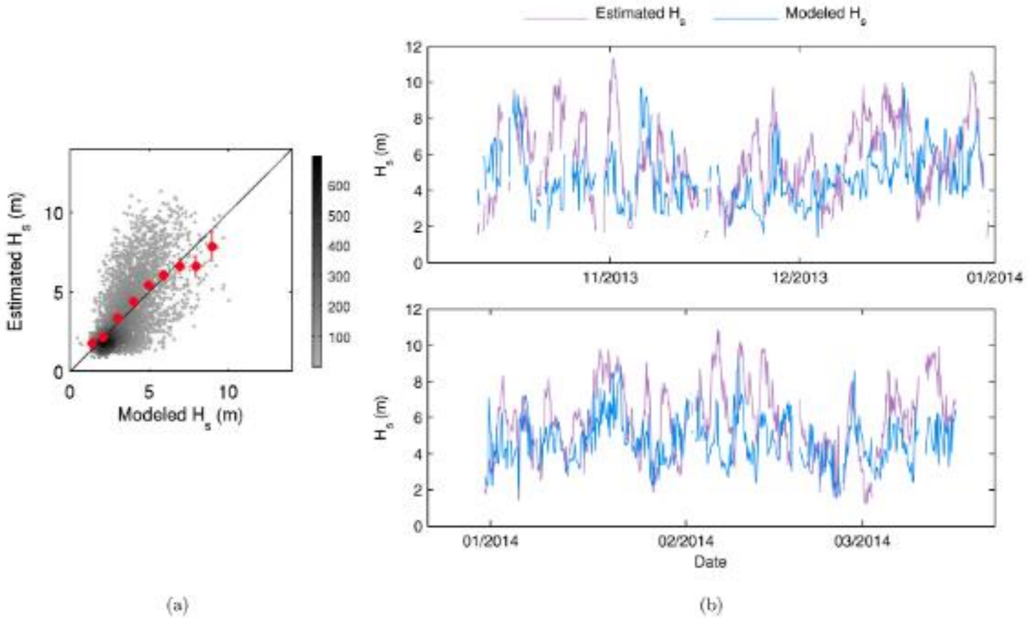
- Study deep structure of the earth
- Track storms far away the seismometers (3000-10000 km)
- Monitor remote ocean wave height



Poli et al., Science 2012

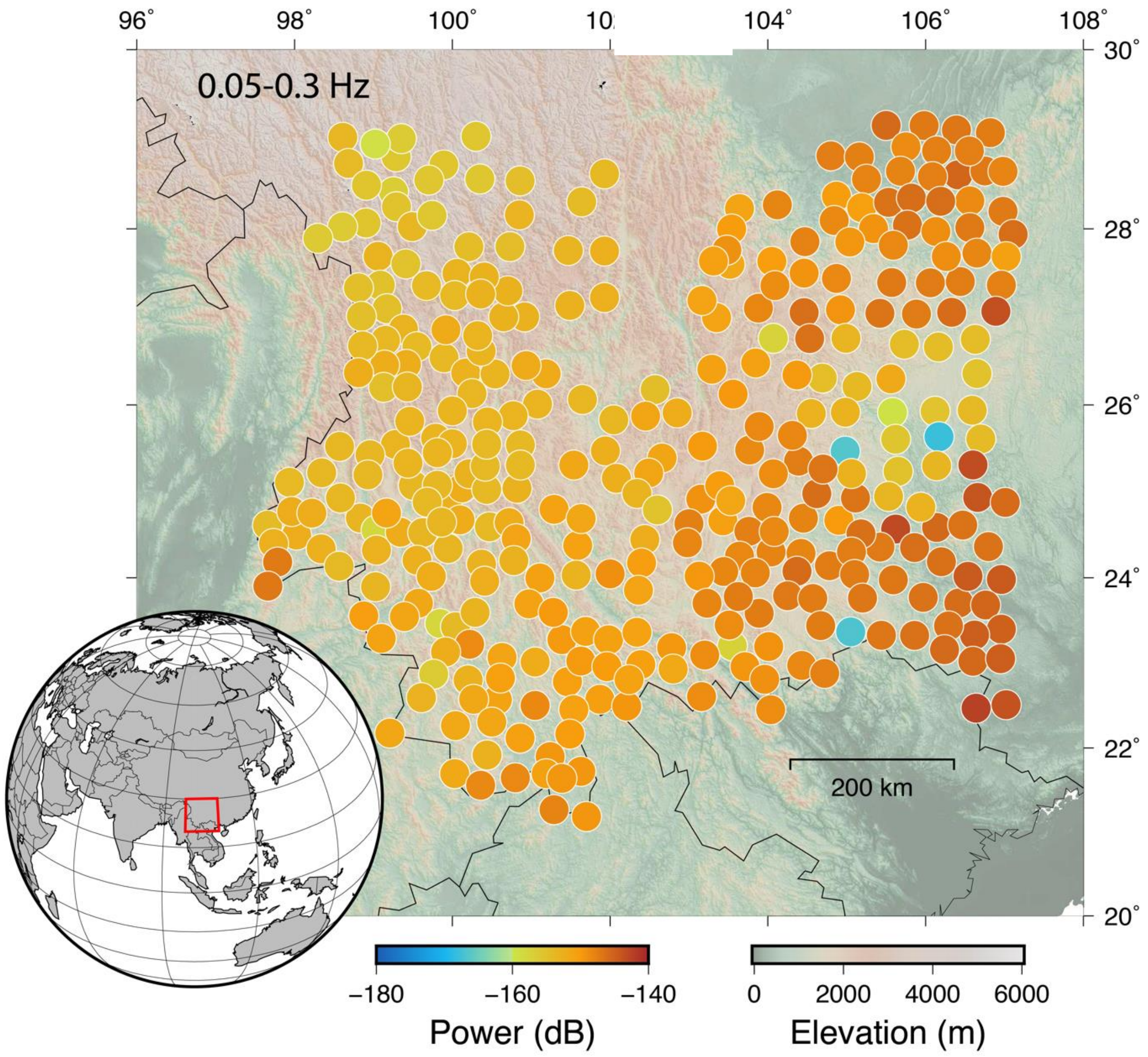


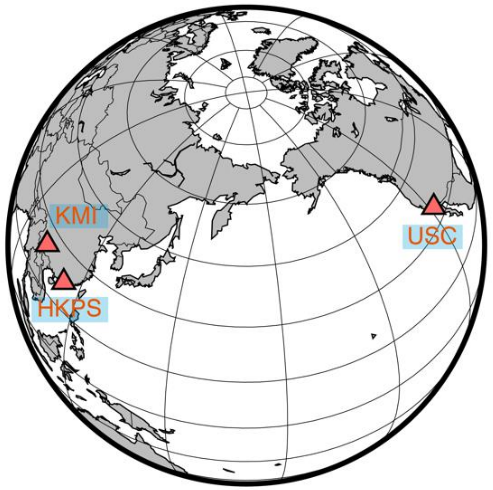
Nishida et al., Science 2016

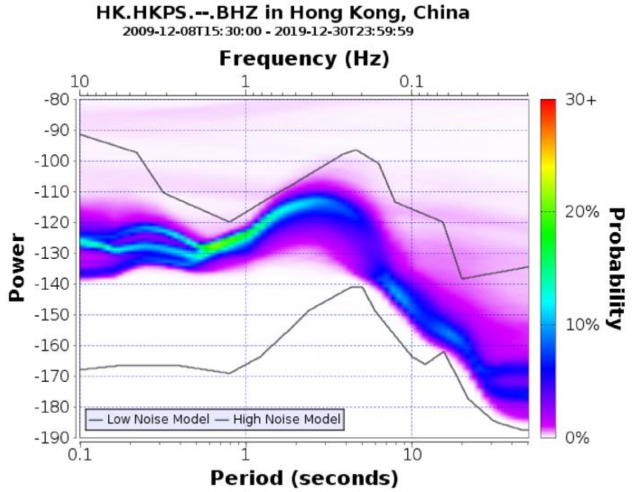


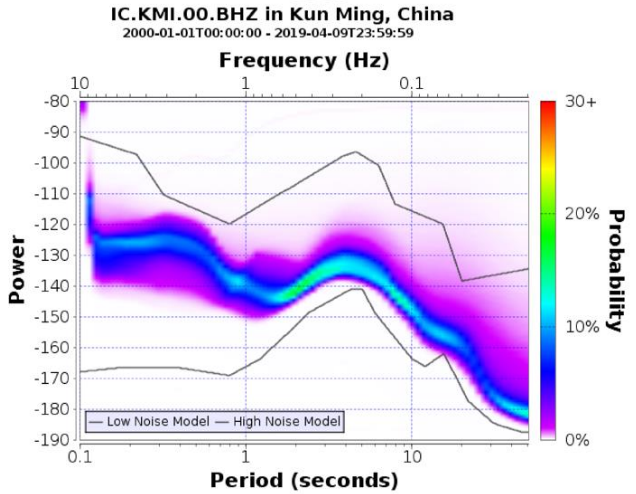
Neale et al., JGR 2016

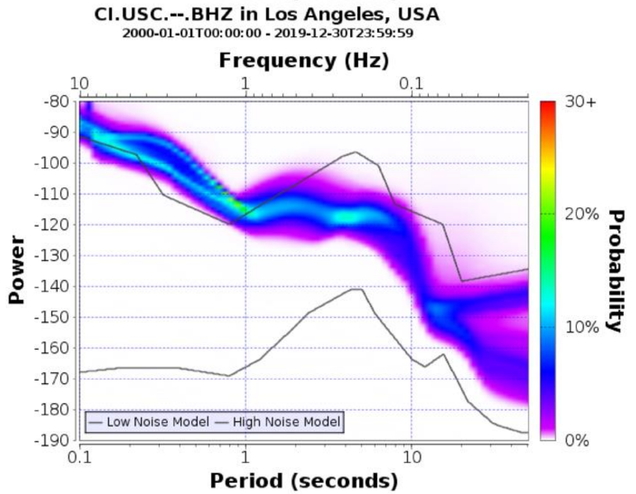
1. Introduction
ChinArray and beamforming example





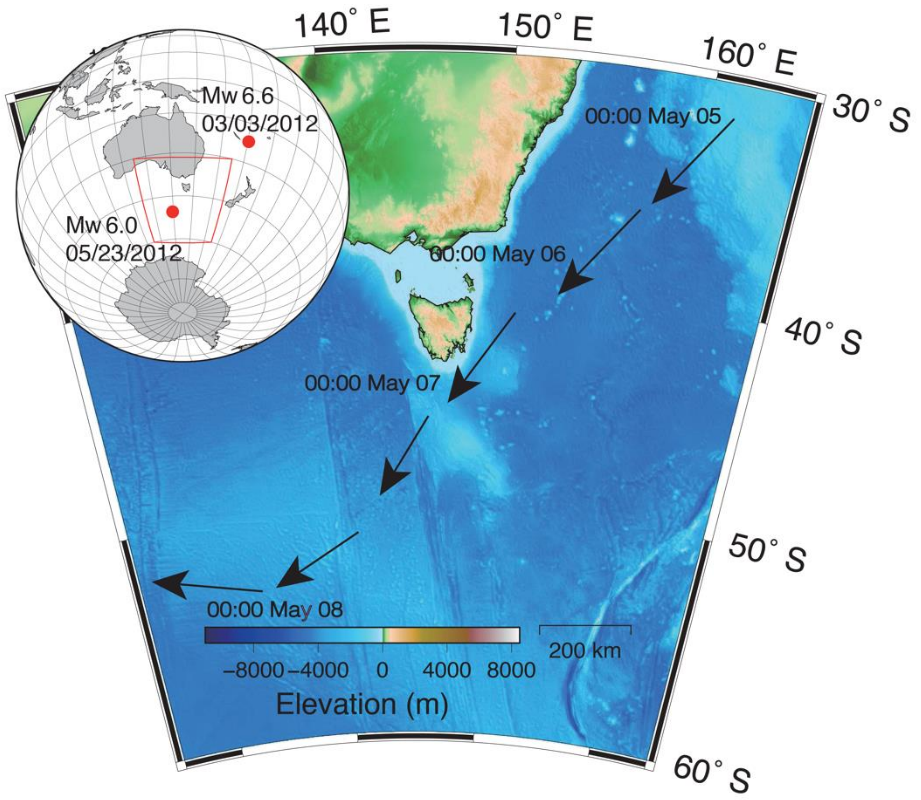


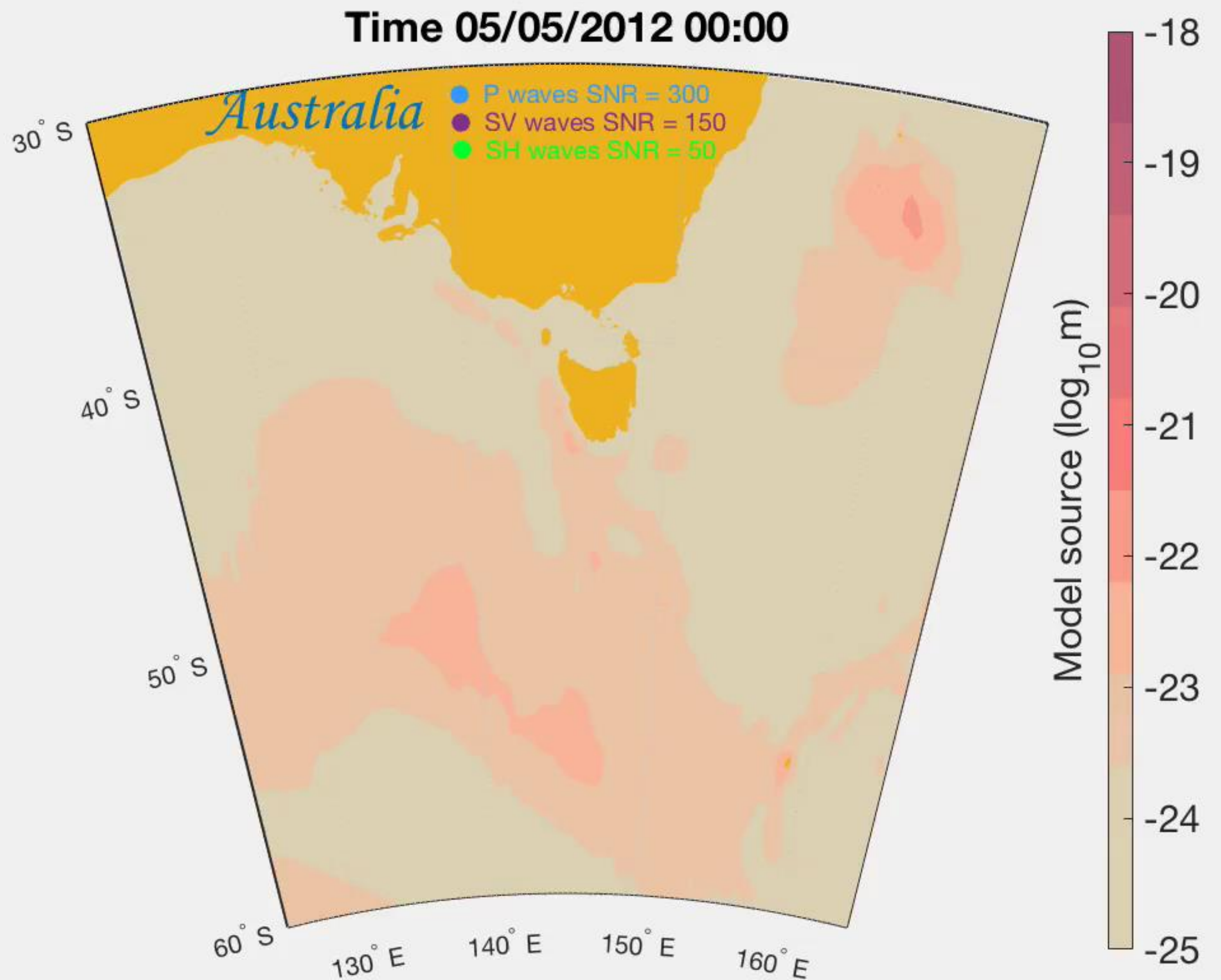




1. Introduction

Storm tracking in southeast Australia





Outline

1. Introduction of seismic noise

2. Machine learning in seismic noise application

3. Traditional denoising methods

4. Machine learning denoising

2. Machine learning in seismic noise application

Machine Learning Basics

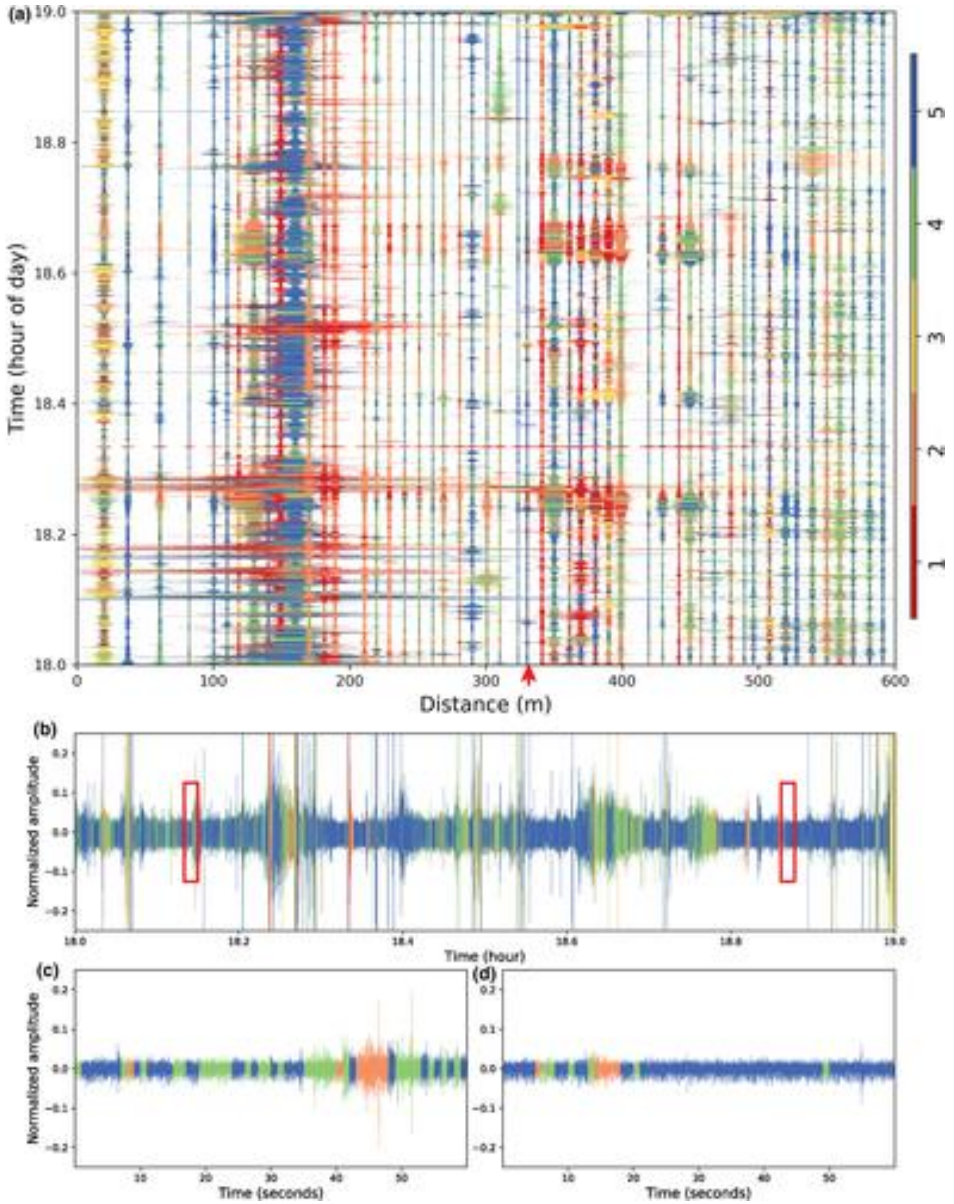
- Supervised learning (requires labeled data, such as classifying earthquake events)
- unsupervised learning (clustering noise and signals)
- self-supervised learning (no label denoising).
- Deep learning: Applications of convolutional neural networks (CNN), recurrent neural networks (RNN), Autoencoder, etc. in earthquake data.

The role of machine learning in seismology:

- Data processing: denoising, signal enhancement, feature extraction.
- Event detection: identifying weak earthquakes or precursor events.
- Big data and the need for automation

2. Machine learning in seismic noise application

Unsupervised Learning



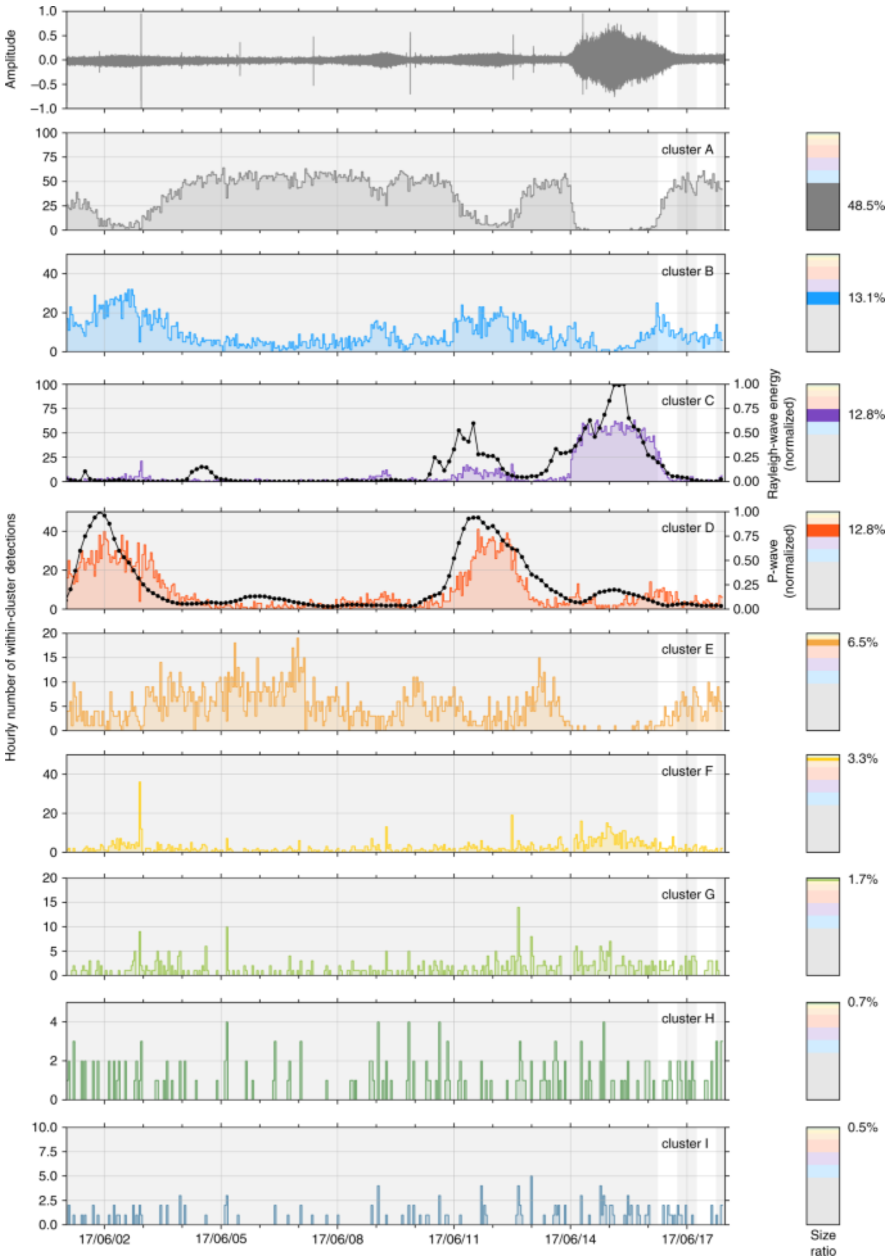
Identifying Different Classes of Seismic Noise Signals Using Unsupervised Learning

- Emergent and impulsive signals in continuous seismic waveforms are identified using cluster analysis on a dense array data
- The unsupervised learning model is trained to identify multiple classes of noise using temporal and spectral data features

Johnson et al., 2019

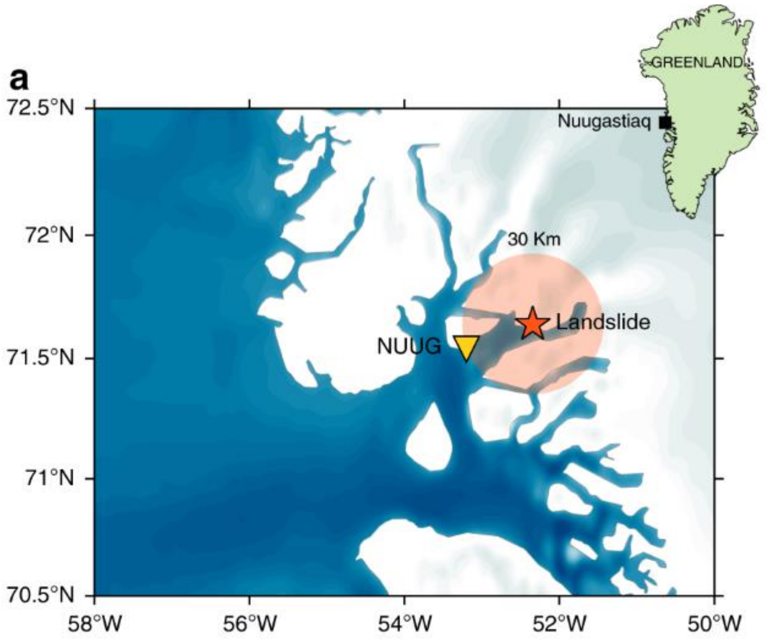
2. Machine learning in seismic noise application

Unsupervised Learning



Clustering earthquake signals and background noises in continuous seismic data with unsupervised deep learning

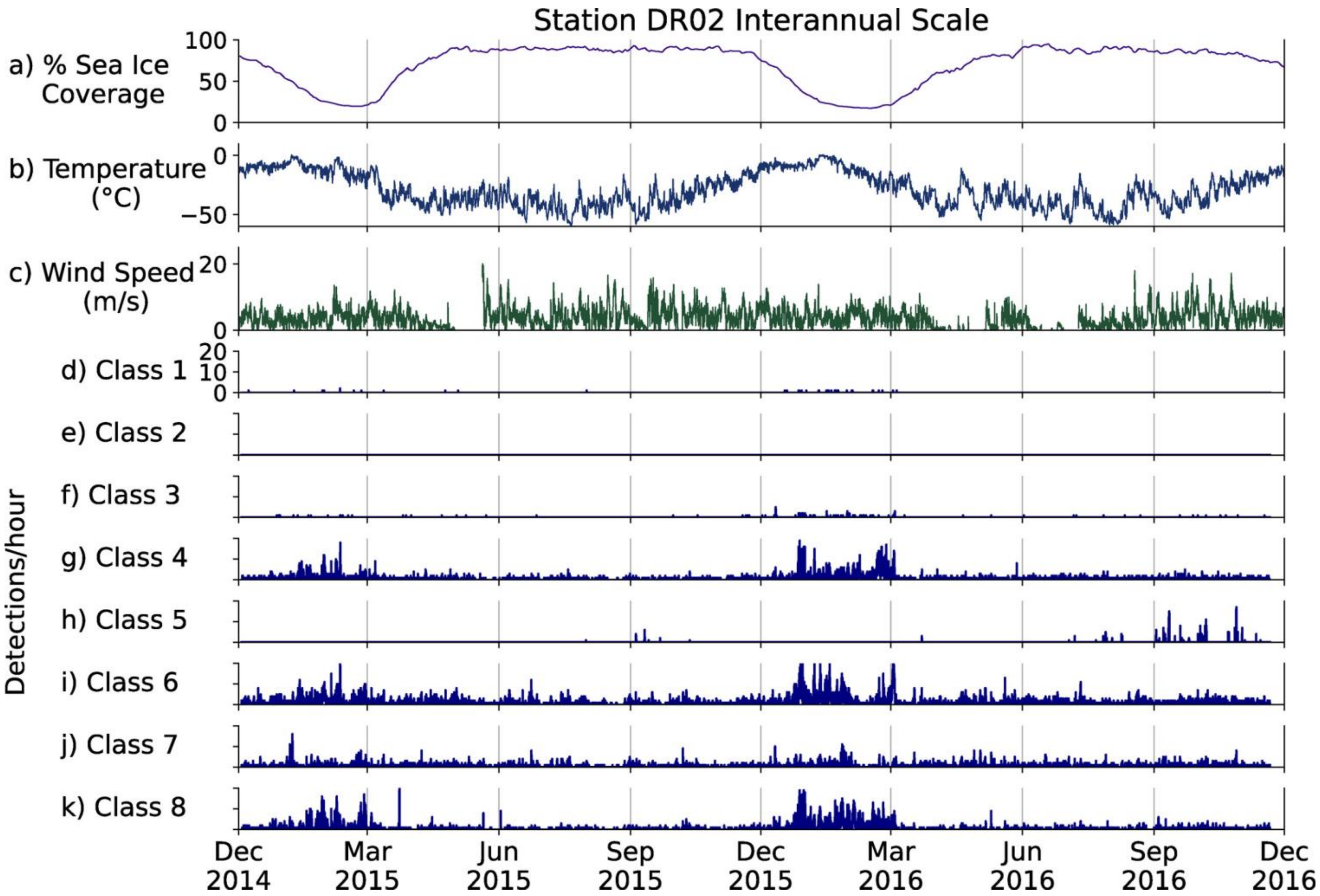
- They demonstrate the blind detection and recovery of the repeating precursory seismicity that was recorded before the main landslide rupture



Seydoux et al., 2020

2. Machine learning in seismic noise application

Unsupervised Learning



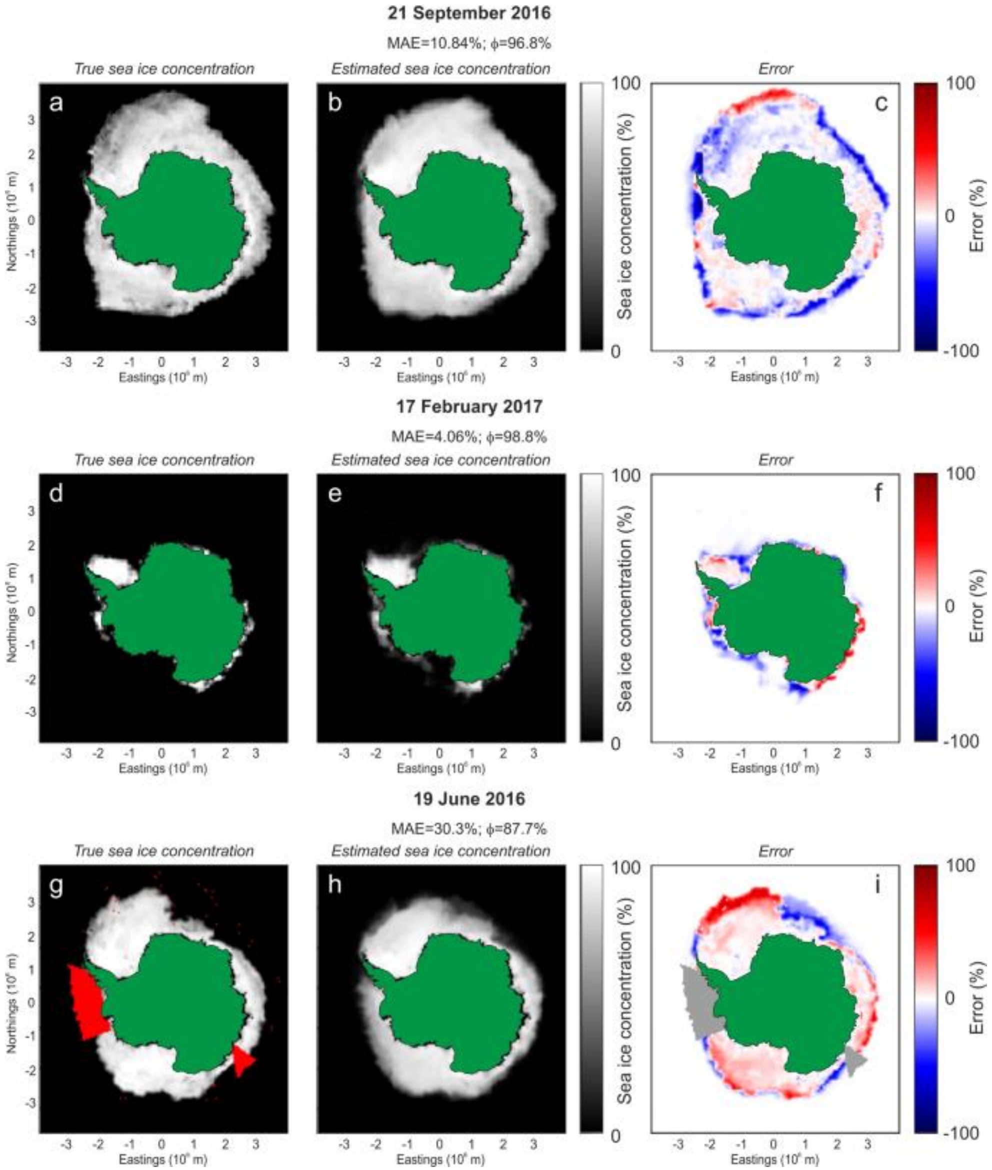
Unsupervised Deep Clustering of Seismic Data: Monitoring the Ross Ice Shelf, Antarctica

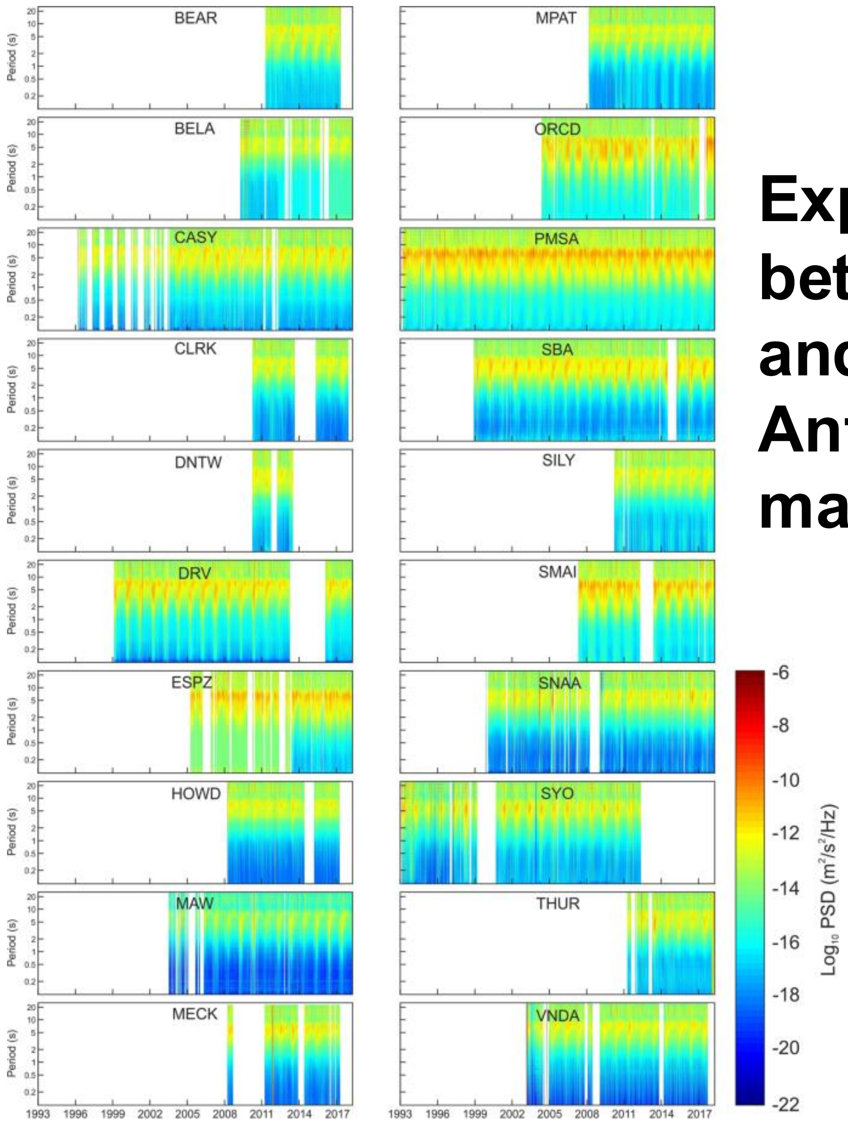
- Eight classes of dominant seismic signals were identified and compared with environmental data such as temperature, wind speed, tides, and sea ice concentration.

Jenkins II et al., 2021

2. Machine learning in seismic noise application

Supervised Learning





Exploring the link between microseism and sea ice in Antarctica by using machine learning

Cannata et al., 2019

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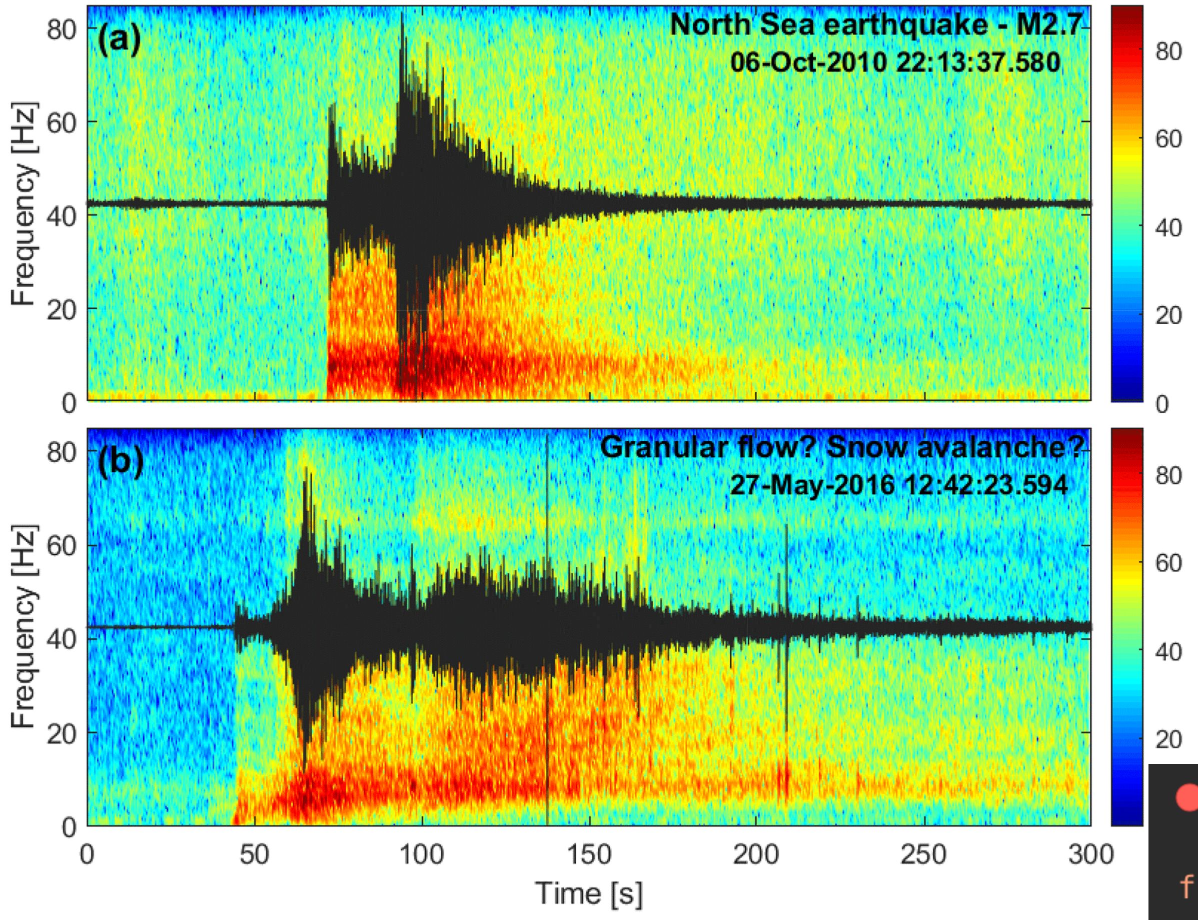
3. Traditional denoising methods



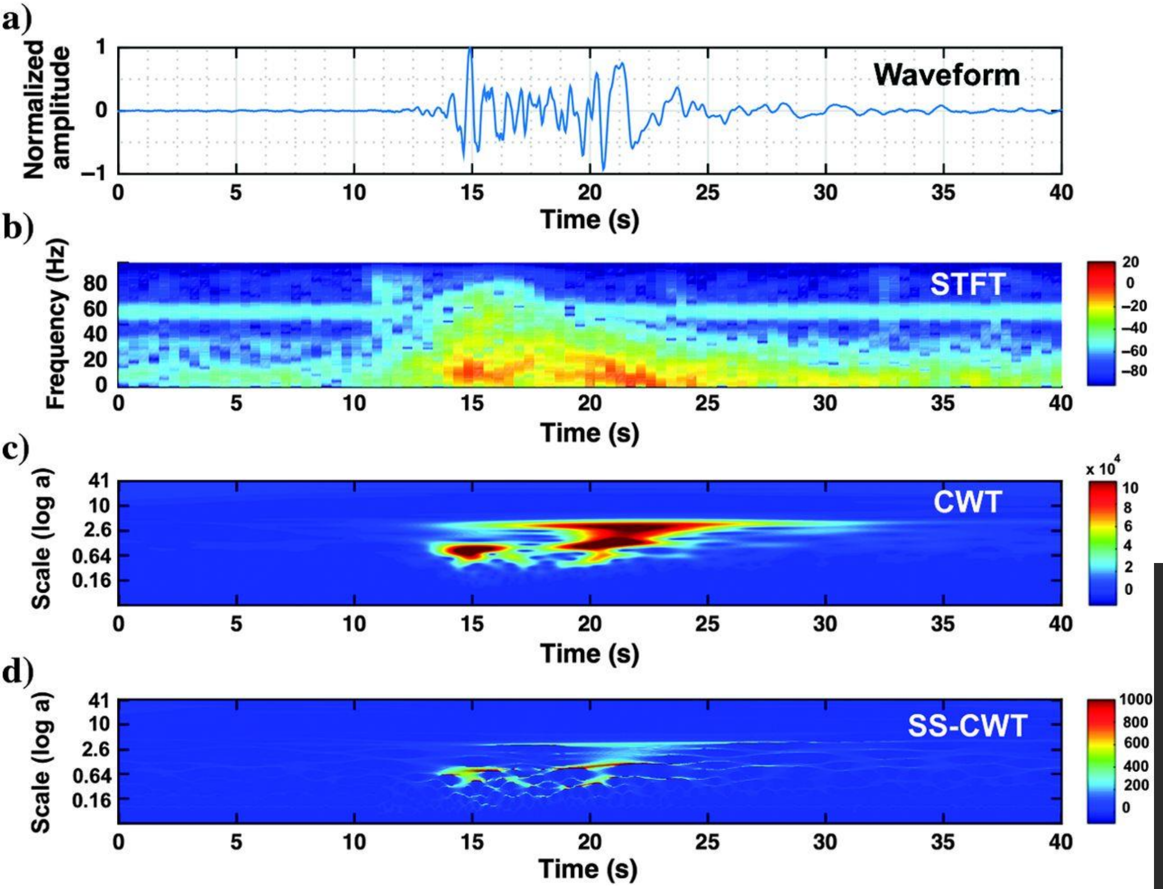
Band-Pass Filter

- View the spectrum through Fourier transform, filter out abnormal frequency components and then perform inverse transform.
- Preliminary processing
- Standard operations before earthquake event
- By retaining a specific frequency band (e.g. 1–20 Hz), low-frequency drift and high-frequency interference are removed

```python
from scipy.signal import butter, filtfilt

b, a = butter(4, [lowcut / (fs / 2), highcut / (fs / 2)], btype='band')
# Apply the filter using zero-phase filtering
filtered = filtfilt(b, a, data)
```

3. Traditional denoising methods



Mousavi et al., 2016

Wavelet Denoising

- Decompose the waveform into multiple scales, remove the noise in the wavelet coefficients by threshold, and then reconstruct
- Applicable to non-stationary and transient seismic signals

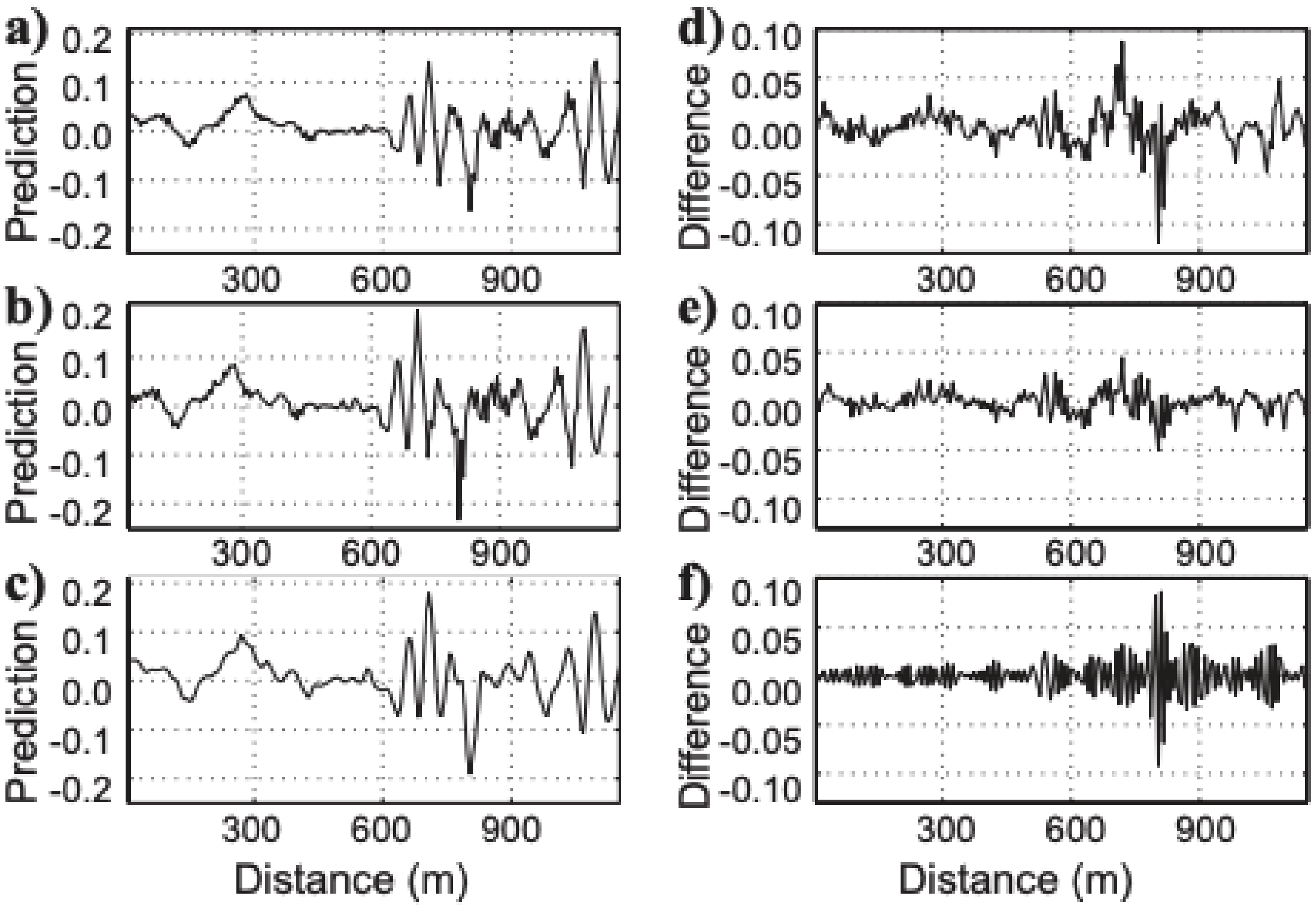
```
import pywt
import numpy as np

def wavelet_denoise(data, wavelet="db4", level=3):
    coeff = pywt.wavedec(data, wavelet, mode="per")
    sigma = np.std(coeff[-level])
    uthresh = sigma * np.sqrt(2 * np.log(len(data)))

    coeff_denoised = [pywt.threshold(c, value=uthresh, mode="soft") if i > 0 else c
                      for i, c in enumerate(coeff)]
    return pywt.waverec(coeff_denoised, wavelet, mode="per")

denoised = wavelet_denoise(data)
```

3. Traditional denoising methods



Bekara and Baan, 2009

Empirical Mode Decomposition (EMD)

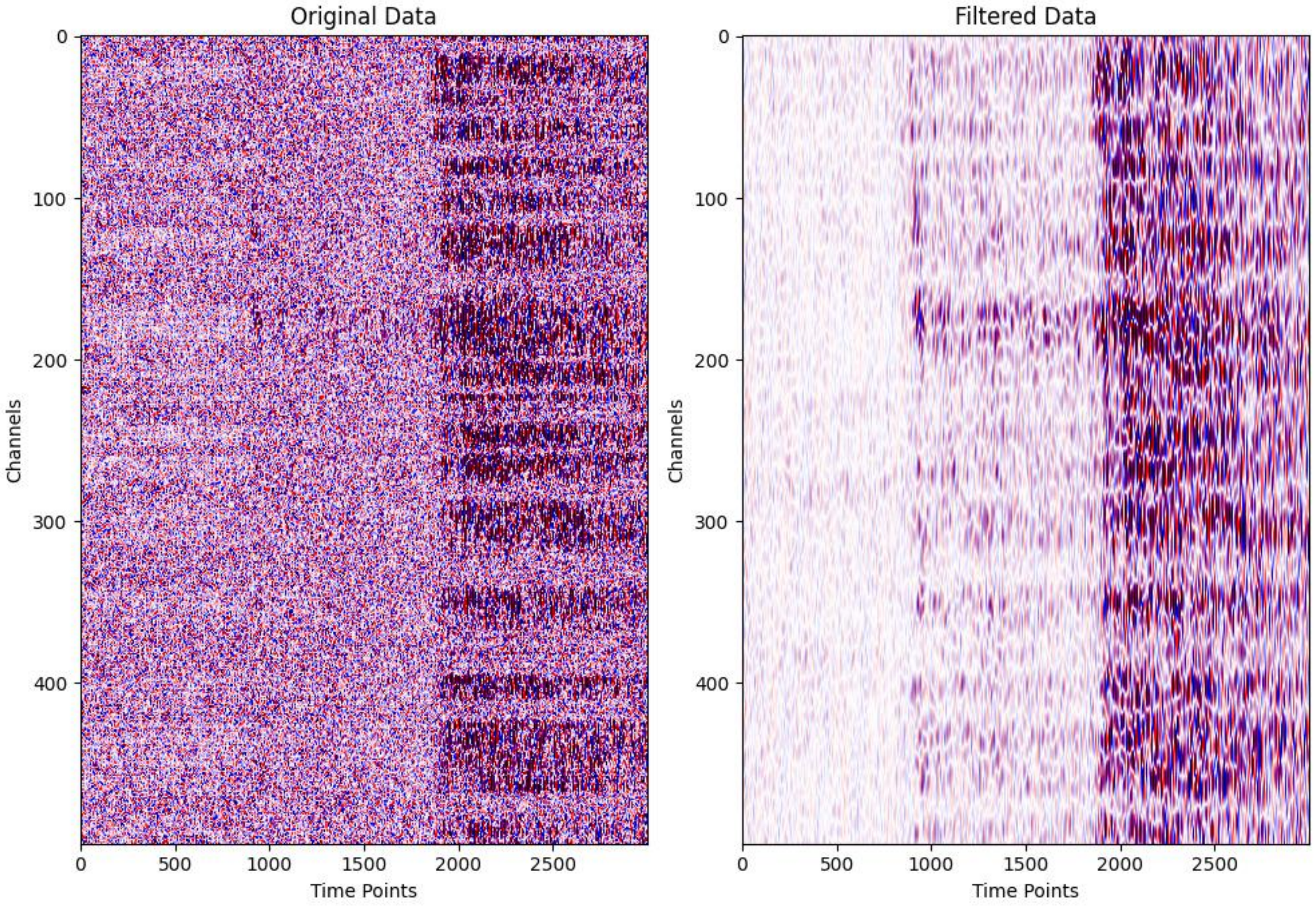
- Decompose the signal into multiple intrinsic mode functions (IMFs), remove high-frequency or low-frequency components and reconstruct them
- Applicable to non-stationary and transient seismic signals

```
import numpy as np
from PyEMD import EMD
from obspy import read

st = read("example.sac")
tr = st[0]
signal = tr.data
fs = tr.stats.sampling_rate
time = np.linspace(0, len(signal)/fs, len(signal))

# Apply Empirical Mode Decomposition (EMD)
emd = EMD()
imfs = emd(signal)
```

3. Traditional denoising methods



f–k filtering

- f–k filtering converts a two-dimensional signal into the frequency-wavenumber domain.
- Distinguishable spatial directions
- Strong velocity separation capability
- Requires two-dimensional data
- Non-adaptive filter design requires manual setting of parameters (fmin/kmax, etc.)
- Edge effect - frequency domain processing may cause artifacts, requiring window function or preprocessing optimization

3. Traditional denoising methods

f–k filtering in Python

```
def velocity_fk_filter(data, fs, dx, v_min, v_max):

    n_channels, n_time_points = data.shape

    dt = 1 / fs  #
    freq = np.fft.fftfreq(n_time_points, dt)
    freq = fftshift(freq)  # Shift zero frequency to the center

    dk = 1 / (n_channels * dx)
    wavenumber = np.fft.fftfreq(n_channels, dx)  # S
    wavenumber = fftshift(wavenumber)

    # Create 2D frequency-wavenumber grid
    f, kx = np.meshgrid(freq, wavenumber)

    # Compute phase velocity (v = f / kx)
    v = np.divide(f, kx, out=np.zeros_like(f), where=kx != 0)  # Avoid division by zero

    # Create FK filter mask
    filter_mask = (v_min <= np.abs(v)) & (np.abs(v) <= v_max)

    # Apply FFT to data
    fk_data = fft2(data)
    fk_data = fftshift(fk_data)  # Shift zero frequency to the center

    # Ensure mask matches fk_data shape
    if filter_mask.shape != fk_data.shape:
        raise ValueError(
            f"Filter mask shape {filter_mask.shape} does not match FK data shape {fk_data.shape}")

    fk_data_filtered = fk_data * filter_mask

    # Transform back to time-space domain
    fk_data_filtered = ifftshift(fk_data_filtered)
    filtered_data = np.real(ifft2(fk_data_filtered))

  return go(f, seed, [])
}
```

Outline

1. Introduction of seismic noise

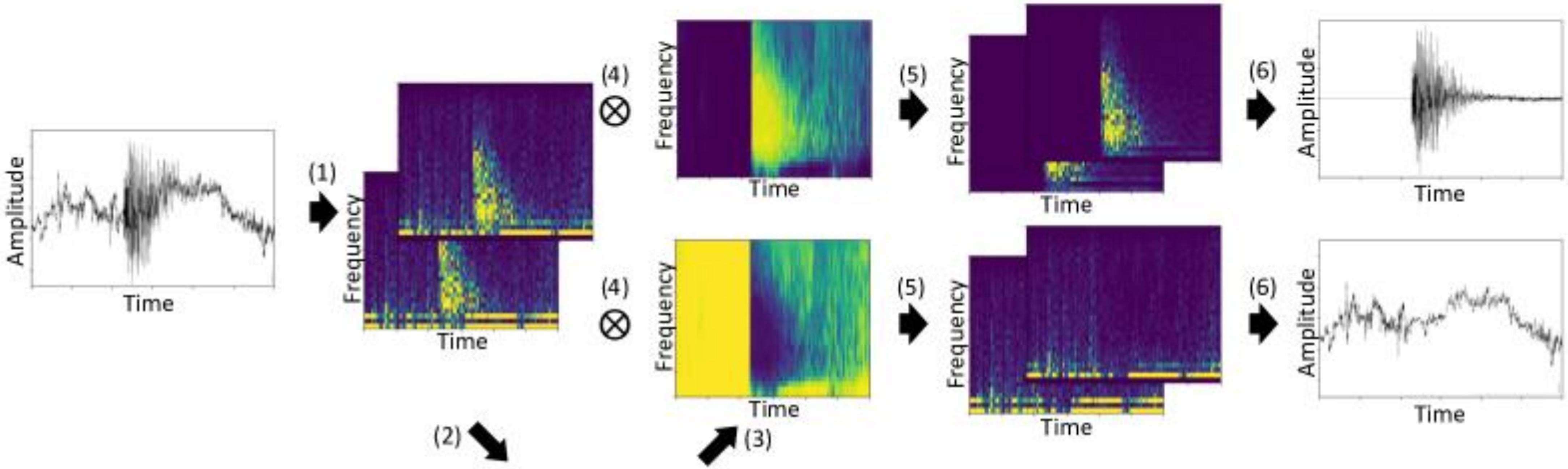
2. Machine learning in seismic noise application

3. Traditional denoising methods

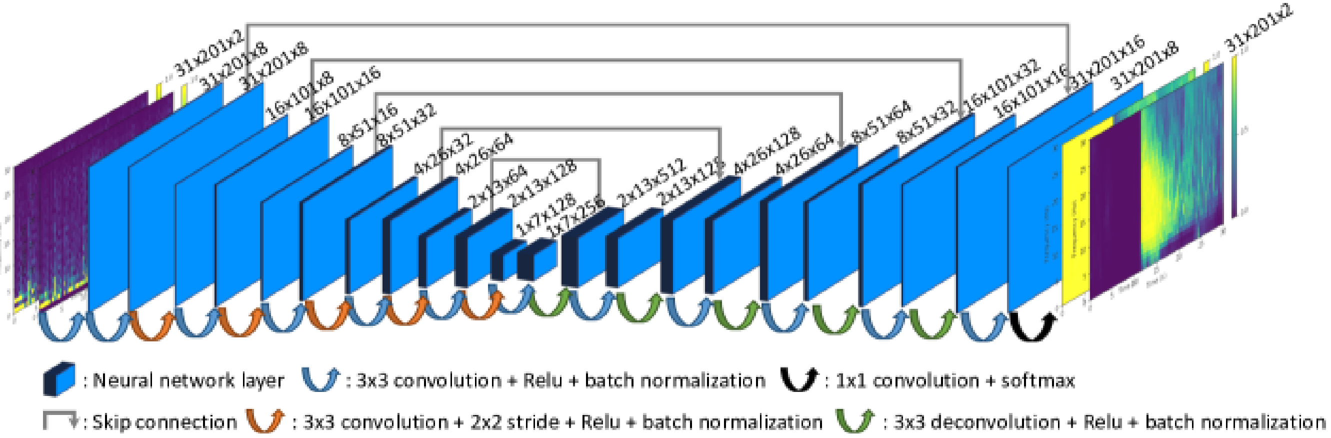
4. Machine learning denoising

4. Machine learning denoising

Supervised Learning



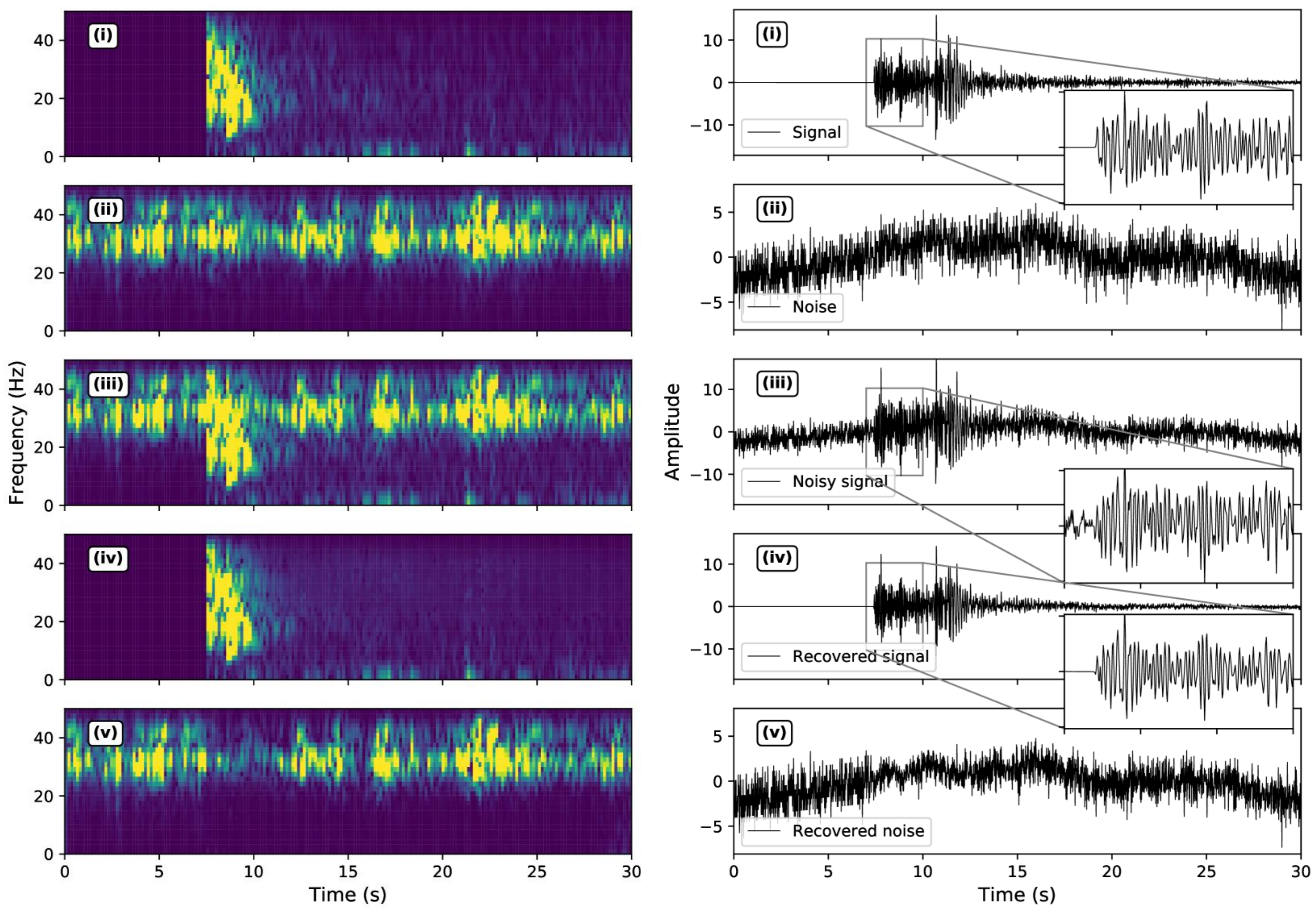
DeepDenoiser



Zhu et al., 2019

4. Machine learning denoising

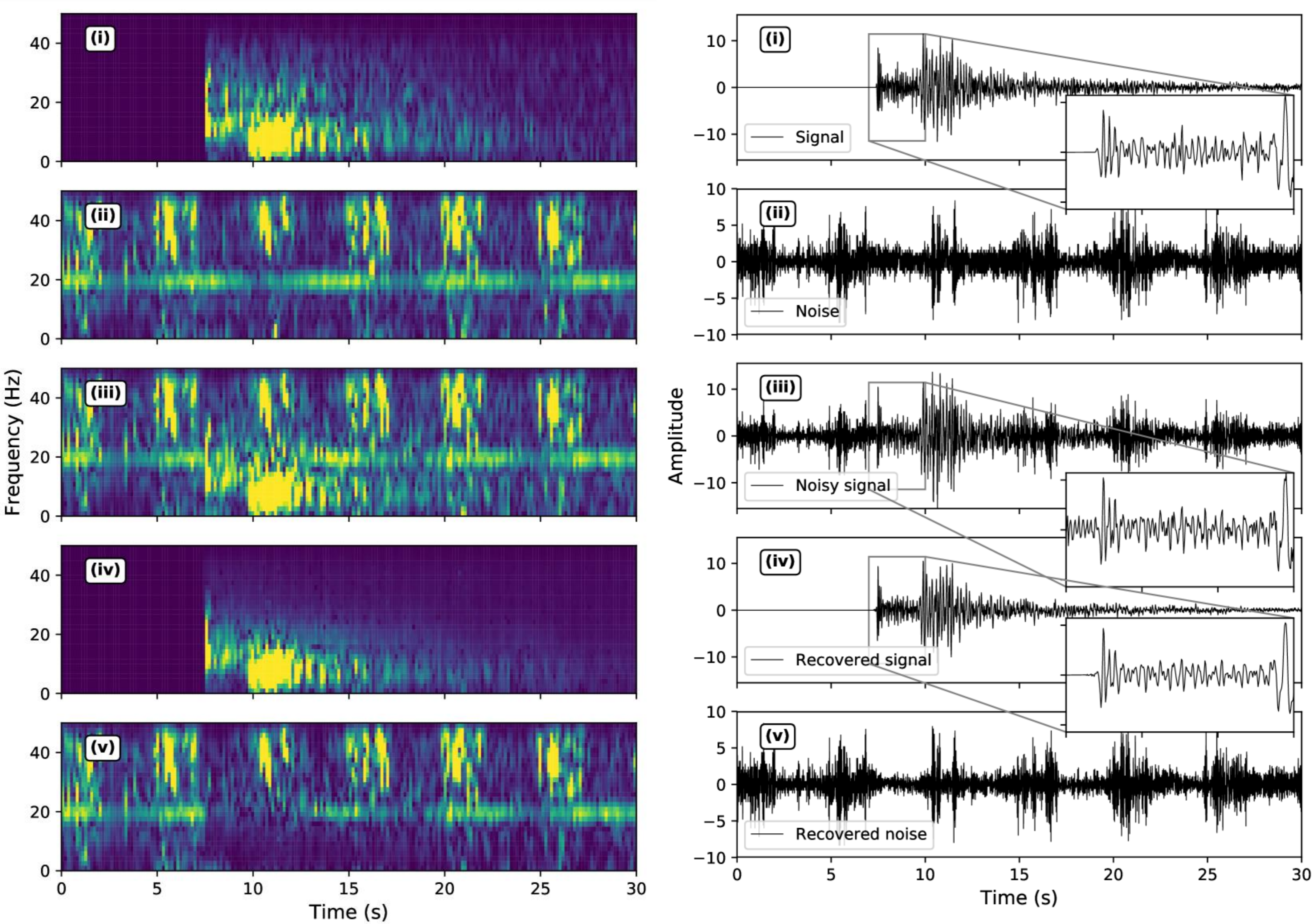
Supervised Learning



Zhu et al., 2019

4. Machine learning denoising

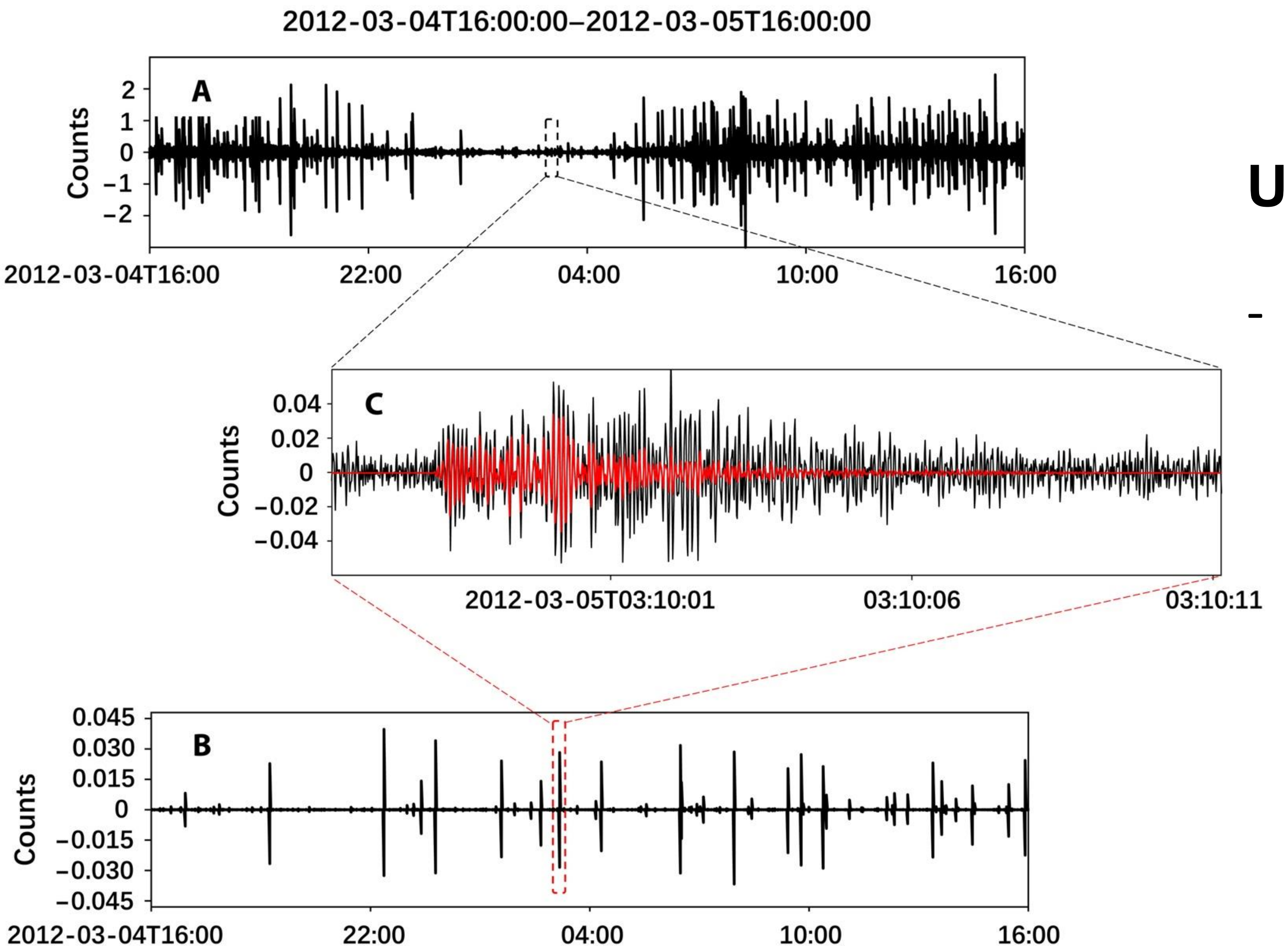
Supervised Learning



Zhu et al., 2019

4. Machine learning denoising

Supervised Learning



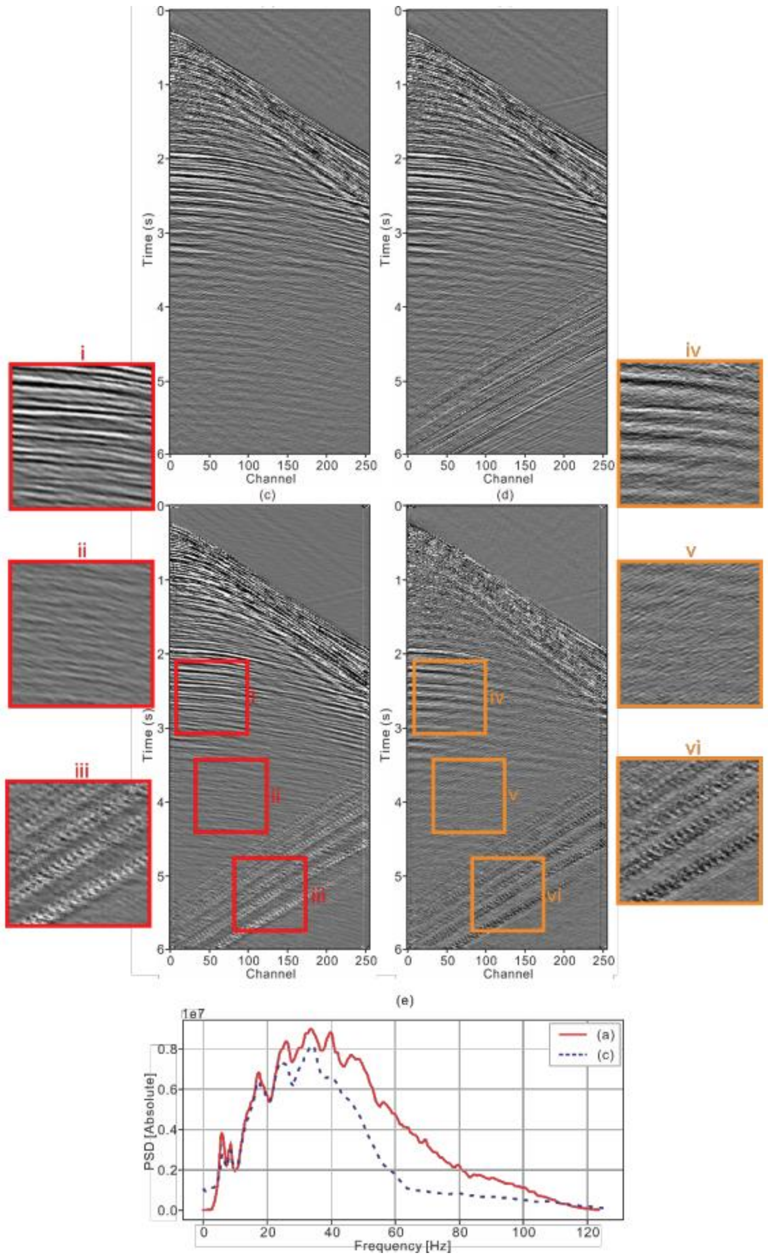
UrbanDenoiser

- Urban earthquake monitoring through deep-learning-based noise suppression

Yang et al., 2022

4. Machine learning denoising

Supervised Learning



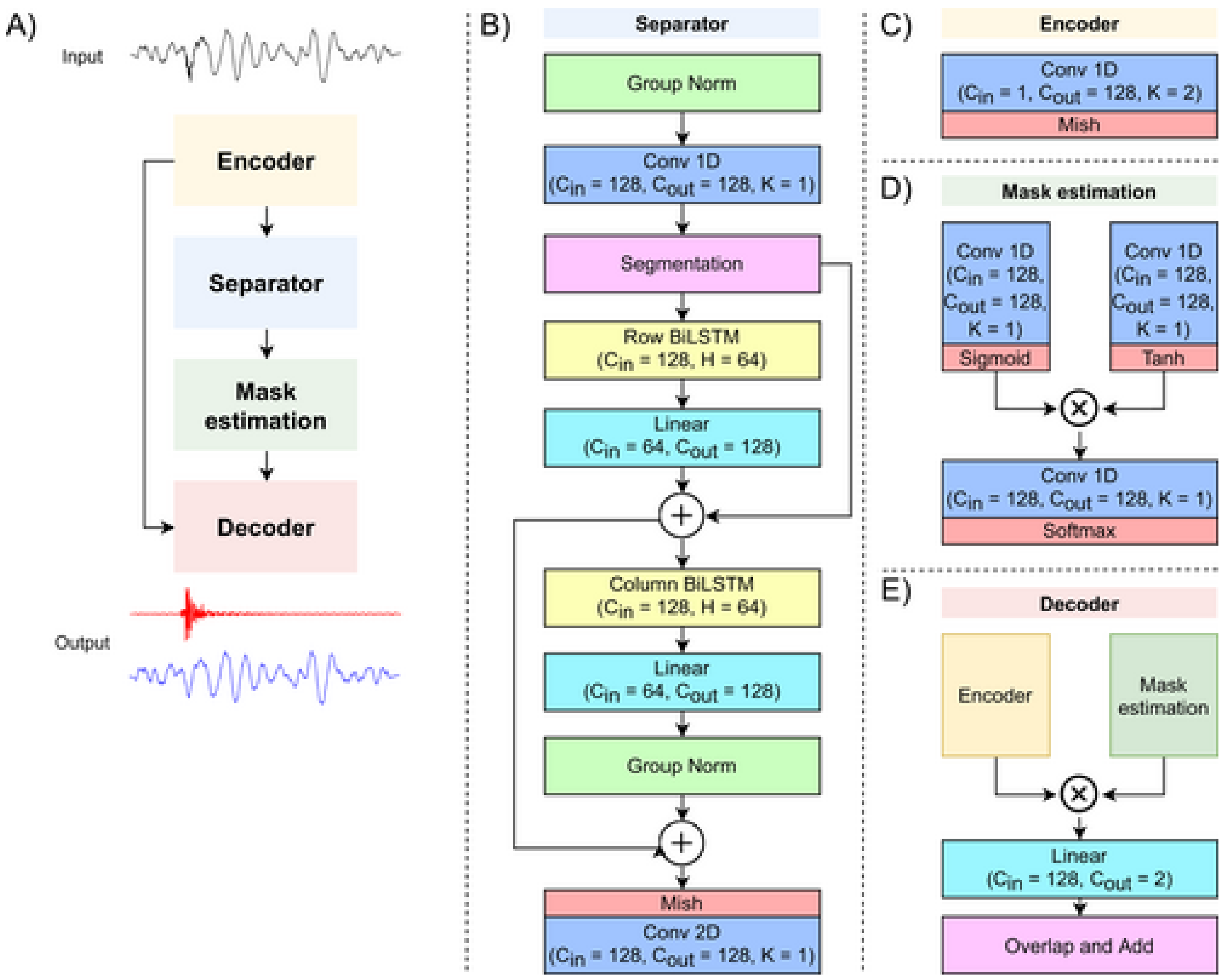
Marine seismic survey denoiser

- Attenuation of marine seismic interference noise employing a customized U-Net

Sun et al., 2022

4. Machine learning denoising

Supervised Learning



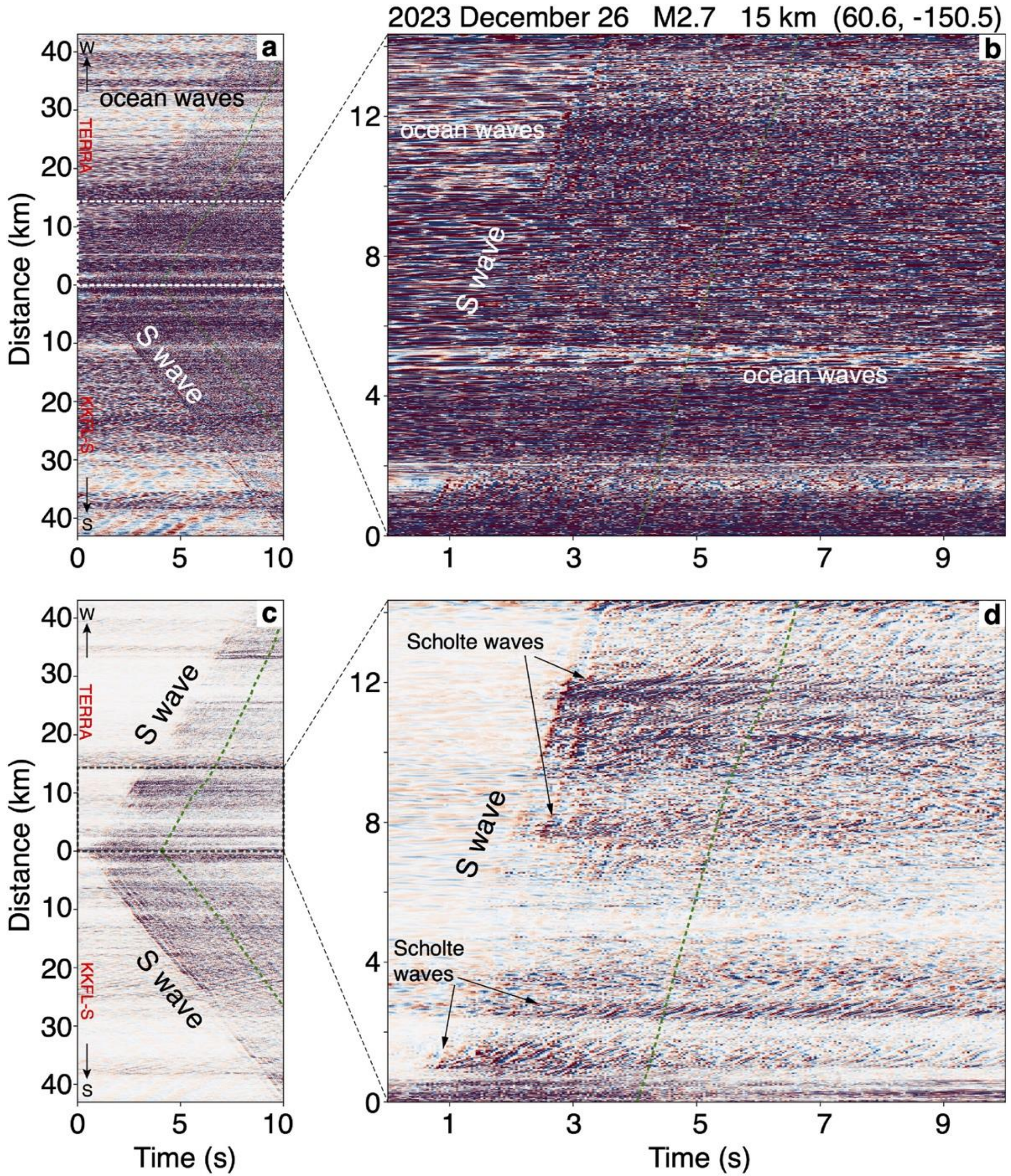
SEDENOSS

- SEparating and DENOising Seismic Signals With Dual-Path Recurrent Neural Network Architecture

Novoselov et al., 2022

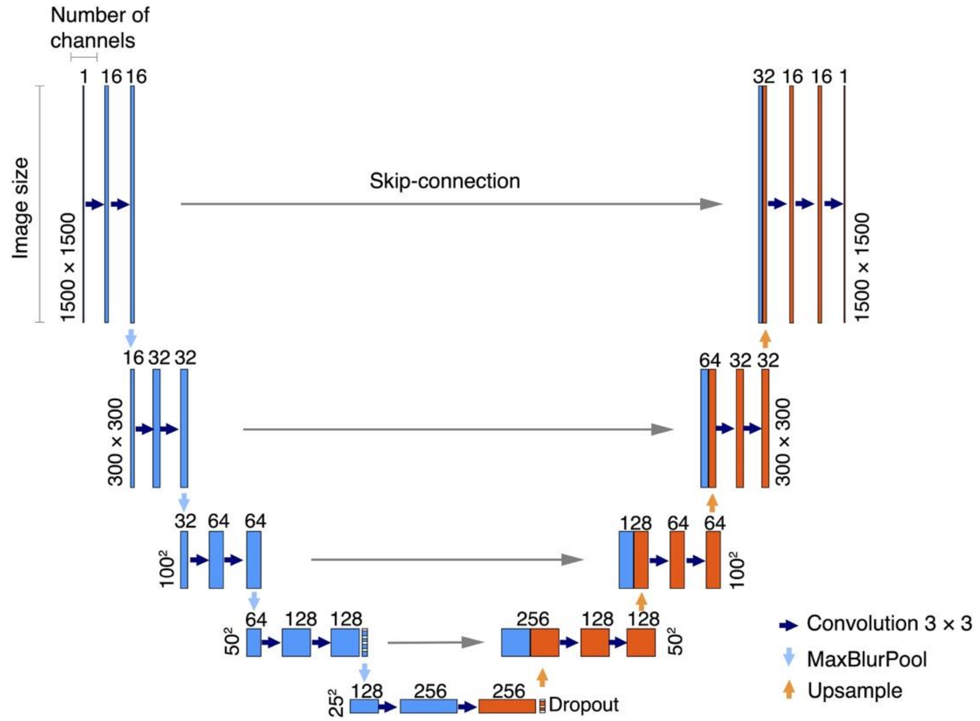
4. Machine learning denoising

self-supervised Learning



Denoising Distributed Acoustic Sensing Using Masked Auto-Encoders

- Self-supervised deep learning algorithm, a masked auto-encoder (MAE), to denoise DAS data



Shi et al., 2025

Future challenges of ML in seismic noise analysis

1. Data Quality and Availability	Limited high-quality labeled datasets for training.
	Difficulty in obtaining pure seismic noise (single source) data across environments.
2. Model Generalization	Models struggle to perform well across varied geological settings or noise types.
	Overfitting to specific datasets reduces robustness.
3. Computational Demands	Deep learning requires significant computational resources.
	Real-time processing for large-scale seismic data is challenging.
4. Interpretability	"Black-box" nature of models hinders trust and adoption in seismology.
	Need for explainable AI to align with physical principles.
5.Integration with Traditional Methods	Combining machine learning with conventional techniques (e.g., f-k filtering) is complex. Balancing data-driven and physics-based approaches.

Thank you!